Chapter 1

Introduction

1.1 Mathematical Modelling of Bacterial Resistance to Multiple Antibiotics and Immune System Response

1.1.1 Background and Motivation

Bacterial infections have historically been among the leading causes of morbidity and mortality worldwide. The discovery of antibiotics transformed once fatal infections into treatable conditions. However the widespread use of antibiotics has led to the emergence of resistant bacterial strains, which now pose a serious threat to global public health.

Antibiotic resistance occurs when bacteria evolve mechanisms that allow them to survive exposure to drugs designed to kill or inhibit them.

Mathematical modelling offers a powerful approach to studying resistance dynamics. By translating biological interactions into equations, models allow researchers to analyze system behavior, identify critical parameters, and evaluate treatment strategies in a controlled and reproducible way. This project applies such an approach, using a system of ordinary differential equations to model the interaction between sensitive bacteria, resistant bacteria, immune cells, and multiple antibiotics within a single host.

1.1.2 Problem Statement

Despite decades of research, bacterial resistance to antibiotics continues to rise at an alarming rate. Clinical treatment decisions such as which antibiotics to use, at what dose, and for how long are often made without a full quantitative understanding of how resistance develops and how the immune system interacts with both the bacteria and the drugs. Most existing mathematical models either ignore the immune response entirely or treat it as a constant background effect, which fails to capture the dynamic and adaptive nature of host immunity.

Furthermore, the majority of models focus on either the spread of resistance across a population or the pharmacokinetics of a single antibiotic, rarely combining multiple antibiotics with a dynamic immune response in a single within-host framework. This leaves a gap in understanding how these factors interact simultaneously during the course of treatment in an individual patient.

This project addresses that gap by implementing and analyzing the within-host model, which incorporates sensitive and resistant bacterial populations, a dynamic immune cell population, and the concentrations of multiple antibiotics simultaneously. The central question is: under what conditions does antibiotic treatment succeed in controlling infection, and what role does immune system strength play in determining that outcome?

1.2 Logistic Population Models with Harvesting

1.2.1 Background and Motivation

The study of population dynamics has been a fundamental concern in both ecology and applied mathematics. Understanding how populations grow, stabilize, and decline is essential for managing natural resources, particularly in fisheries where human harvesting directly influences population behavior.

The earliest formal mathematical treatment of population growth was proposed by Malthus in 1798, who described growth as purely exponential. While this captured the basic idea of biological reproduction, it failed to account for environmental limits such as food supply and space. To address this, Verhulst introduced the logistic growth model in 1838, incorporating the concept of carrying capacity i.e. the maximum population size an environment can sustainably support. The logistic model became the foundation for much of modern population ecology.

In fishery management, the logistic model was further developed to include harvesting. Two major strategies have been widely studied: constant harvesting, where a fixed number of fish are removed per unit time, and periodic harvesting, which reflects the seasonal nature of real fishing activity where grounds are opened and closed at regular intervals.

These mathematical models, expressed as ordinary differential equations, allow researchers and policymakers to analyze long-term population behavior, identify critical thresholds, and design sustainable harvesting strategies that prevent extinction while maximizing yield.

1.2.2 Problem Statement

Fish populations worldwide are under increasing pressure from commercial and subsistence fishing. According to the Food and Agriculture Organization of the United Nations, a significant portion of global fish stocks are either fully exploited or overexploited, meaning they are being harvested at or beyond their biological limits. When harvesting exceeds the natural capacity of a population to recover, the population enters a state of irreversible decline that can ultimately lead to commercial extinction or even biological extinction of the species.

The central problem addressed in this study is the determination of safe and sustainable harvesting rates for a fish population governed by logistic growth dynamics. Specifically, the study asks: at what harvesting rate does a fish population remain stable, and at what point does harvesting cause the population to collapse? Furthermore, how does the pattern of harvesting whether constant or periodic affect the long-term survival and stability of the population?

These questions cannot be answered by intuition or observation alone. A rigorous mathematical framework is needed to model the interaction between natural population growth and human harvesting activity, to identify equilibrium states, to assess their stability, and to determine the critical harvesting threshold known as the Maximum Sustainable Yield (MSY) which means the largest harvesting rate that a population can withstand without being driven toward extinction.

1.3 Objectives of the Study

The objectives of this project are as follows:

- To implement the mathematical model numerically using MATLAB and reproduce and verify the analytical equilibrium results through simulation.
- To investigate the effect of immune system strength on treatment outcome and analyze how the fitness cost of resistance influences the persistence of resistant bacteria.
- To evaluate the consequences of early antibiotic termination on bacterial load and compare single antibiotic therapy with combination therapy quantitatively.
- To perform a sensitivity analysis identifying the most critical model parameters and investigate the sensitivity of the model to changes in key parameters including the growth rate k, carrying capacity N, and harvesting rate a.

- To derive and analyze the logistic growth model with constant harvesting and determine its equilibrium points using the quadratic formula and classify the stability of each equilibrium point using linearization and phase portrait analysis.
- To derive the Maximum Sustainable Yield (MSY) and determine the critical harvesting threshold beyond which population extinction occurs and to formulate and numerically analyze the logistic growth model with periodic harvesting using the Runge-Kutta method (ode45).
- To compare the long-term behavior of fish populations under constant and periodic harvesting strategies through graphical and numerical simulation.

1.4 Scope and Limitations

This study operates within the following scope and is subject to the following limitations:

Scope:

- This study focuses exclusively on within-host dynamics, the behavior of bacteria, immune cells, and antibiotics within a single infected individual, as well as single-species fish population models governed by the logistic differential equation with two forms of harvesting which are constant and periodic.
- The model considers two antibiotics administered simultaneously as combination therapy.
- The study considers a closed fish population with no immigration or emigration and analyzes two harvesting models: constant harvesting and sinusoidal periodic harvesting.
- Analysis covers both mathematical stability and numerical simulation over a 90-day treatment period for the bacterial model and across a defined range of parameter values for the population model with MATLAB serving as the primary computational tool for numerical integration and graphical visualization in both models.

Limitations:

- The bacterial model does not capture the spread of infection between individuals or across hospital populations, and resistance is assumed to arise solely through spontaneous mutation, horizontal gene transfer and conjugation between bacteria are not modelled.
- The immune response is represented by a single population of cells growing logistically in proportion to bacterial load, innate immunity, antibody responses, and cytokine signaling are not included.
- The population model assumes a single isolated species with no predator-prey interactions or competition from other species, and environmental factors such as temperature, disease, pollution, and habitat loss are not incorporated into the population model.
- Both models treat population size as a continuous variable, which is a mathematical idealization but in reality, population size is a discrete whole number.

1.5 MATLAB

MATLAB (Matrix Laboratory) is a high-level programming language and numerical computing environment developed by MathWorks. It is widely used in engineering, mathematics, and science for numerical analysis, matrix computation, algorithm development, and data visualization. Unlike general purpose programming languages, MATLAB is designed specifically for mathematical computation, making it particularly well suited for solving and visualizing systems of differential equations.

In this project MATLAB was used for the following purposes:

- **Numerical Simulation**: The five equation ODE system was implemented as a MATLAB function and solved using the built-in solver ode45, which uses an adaptive Runge-Kutta method to integrate the system.

- **Parameter Management**: All model parameters were organized into a struct object, allowing different simulation cases to be run by changing a single parameter set without modifying the model function

- **Stability Verification**: Threshold parameters A, B and C were computed numerically and compared against analytical predictions to verify correctness/
- **Visualization**: Time series plots, phase portraits, and comparison plots were generated using MATLABs built in plotting functions to illustrate system dynamics across all simulation scenarios
- **Sensitivity Analysis**: For loops were used to automate parameter sweeps across multiple values of η, c, and treatment duration, producing the comparison plots.
- **Results Export**: Numerical results were organized into formatted tables using MATLAB's table function for inclusion in this report

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Chapter 2

Mathematical Modelling of Bacterial Resistance to Multiple Antibiotics and Immune System Response

2.1 Overview

2.1.1 Bacterial Infections and Antibiotic Resistance

Bacterial infections remain a leading cause of disease and death globally, affecting millions of people each year across all age groups and healthcare settings. The introduction of penicillin in the 1940s marked the beginning of the antibiotic era, offering effective treatment for infections that had previously carried high mortality. Since then, numerous classes of antibiotics have been developed, each targeting different aspects of bacterial physiology such as cell wall synthesis, protein production, or DNA replication.

However, bacteria are capable of rapid evolutionary adaptation. Resistance to antimicrobial agents is now recognized as both a logical and inevitable consequence of antibiotic use. Each new class of antibiotic introduced into clinical practice has been followed within years, and sometimes months, by the emergence of resistant strains. Resistance mechanisms include enzymatic inactivation of the drug, modification of the drug target, active efflux of the drug from the bacterial cell, and reduced membrane permeability. These mechanisms can arise through spontaneous chromosomal mutations or through the acquisition of resistance genes from other bacteria via horizontal gene transfer.

2.1.2 Multiple Antibiotic Therapy

The use of two or more antibiotics simultaneously, known as combination or multiple antibiotic therapy, has become standard practice for several types of bacterial infection. The rationale for this approach is threefold. First, different antibiotics act on different bacterial targets, so their effects can be additive or synergistic, producing greater bacterial killing than either drug alone. Second, combination therapy reduces the probability of resistance emerging, since a single mutation is unlikely to confer resistance to two drugs with different mechanisms simultaneously. Third, for mixed infections involving multiple bacterial species or strains with different susceptibility profiles, combinations ensure that all components of the infection are covered.

2.1.3 Mathematical Modelling in Infectious Diseases

Mathematical modelling has a long history in the study of infectious diseases. The foundational SIR model, which divides a population into susceptible, infected, and recovered compartments, was introduced in the early twentieth century and remains widely used today for studying epidemic dynamics. More sophisticated extensions of this framework have been developed to account for latent infection, vaccination, age structure, spatial heterogeneity, and many other biological complexities

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2.1.4 Within-Host Models vs Epidemiological Models

Mathematical models of infectious disease can be broadly divided into two categories based on their scale of analysis. Epidemiological models, also called between-host models, describe the spread of infection across a population of individuals. These models are used to predict outbreak trajectories, evaluate vaccination programs, and assess the impact of public health interventions. The SIR model and its variants fall into this category.

Within-host models, by contrast, describe the dynamics of infection inside a single individual. These models track the populations of pathogens, immune cells, and drugs over time within one host and are used to understand the mechanisms of infection progression, immune evasion, and treatment response. Within-host models are particularly relevant for studying antibiotic resistance because resistance develops at the level of individual bacterial cells within a patient, driven by mutation and selection pressure from drugs and immunity.

2.2 Mathematical Model

2.2.1 Model Description and Assumptions

This study examines the within-host dynamics of a bacterial infection in a single patient receiving multiple antibiotic treatment. Unlike epidemiological models such as SIR which track disease spread across a population, this model focuses on what happens inside one individual's body. Specifically, how sensitive bacteria, resistant bacteria, immune cells, and antibiotic concentrations interact and change over time.

The model is built as a system of ordinary differential equations (ODEs) with five state variables: the concentration of antibiotic-sensitive bacteria $s(t)$, antibiotic-resistant bacteria $r(t)$, host immune cells $b(t)$, and the concentrations of two antibiotics: isoniazid INH denoted $a_1(t)$ and pyrazinamide PZA denoted $a_2(t)$. All five variables are normalized, meaning their values fall between 0 and 1, where 1 represents the maximum possible concentration or population size.

2.2.2 State Variables and Parameters

State Variables

The model consists of five state variables, all of which are normalized to dimensionless values in the range [0, 1]. Normalization is achieved by dividing each variable by its respective maximum capacity, which simplifies the mathematical analysis and ensures all variables remain biologically meaningful. The following table summarizes the five state variables, their biological meanings, normalizations, and ranges:

Table 2.1: Definition and Normalization of State Variables

Variable	Meaning	Normalization	Range
$s(t)$	Population of antibiotic-sensitive bacteria at time t	$S(t)$ divided by carrying capacity T	[0, 1]
$r(t)$	Population of antibiotic-resistant bacteria at time t	$R(t)$ divided by carrying capacity T	[0, 1]
$b(t)$	Population of host immune cells at time t	$B(t)$ divided by ωT	[0, 1]
$a_1(t)$	Concentration of INH (Isoniazid) at time t	$A_1(t)$ divided by δ_1/μ_1	[0, 1]
$a_2(t)$	Concentration of PZA (Pyrazinamide) at time t	$A_2(t)$ divided by δ_2/μ_2	[0, 1]

Where T is the carrying capacity of bacteria, ω is the ratio of immune cell capacity to bacterial capacity, δ_i is the supply rate of the i^{th} antibiotic, and μ_i is the clearance rate of the i^{th} antibiotic.

Parameters

The model contains the following parameters. Each parameter is strictly positive as required for biological validity. The following table summarizes the model parameters along with their symbols, biological meanings, and units:

Table 2.2: Definition of Model Parameters and Their Units

Parameter	Symbol	Meaning	Unit
Sensitive bacteria growth rate	β_S	Maximum per capita growth rate of sensitive bacteria	day^{-1}
Resistant bacteria growth rate	β_R	Maximum per capita growth rate of resistant bacteria	day^{-1}
Fitness cost of resistance	c	Fractional reduction in growth rate due to carrying resistance genes	dimensionless
Immune killing rate	η	Rate at which immune cells kill bacteria per unit concentration	day^{-1}
Immune cell growth rate	k	Per capita proliferation rate of immune cells in response to infection	day^{-1}
Mutation rate of INH	α_1	Rate at which sensitive bacteria mutate to resistant due to INH exposure	day^{-1}
Mutation rate of PZA	α_2	Rate at which sensitive bacteria mutate to resistant due to PZA exposure	day^{-1}
Antibiotic killing rate of INH	d_1	Rate at which INH kills sensitive bacteria	day^{-1}
Antibiotic killing rate of PZA	d_2	Rate at which PZA kills sensitive bacteria	day^{-1}
INH clearance rate	μ_1	Rate at which INH is cleared from the body	day^{-1}
PZA clearance rate	μ_2	Rate at which PZA is cleared from the body	day^{-1}

2.2.3 The Ordinary Differential Equation System

The model is governed by a system of five ordinary differential equations describing the time evolution of sensitive bacteria, resistant bacteria, immune cells, and two antibiotic concentrations.

Differential Equation for Sensitive Bacteria

$$\frac{ds}{dt} = \beta_S\, s\,(1 - s - r) - \eta\, s\, b - s\,[(\alpha_1 + d_1)a_1 \;+\; (\alpha_2 + d_2)a_2] \qquad \textbf{(2.1)}$$

The following table describes each term in the differential equation governing Sensitive Bacteria:

Table 2.3: Breakdown of the Differential Equation for Sensitive Bacteria

Term	Expression	Meaning
Logistic growth	$\beta_S\, s(1 - s - r)$	Sensitive bacteria grow at rate β_S but slow down as total bacteria s+r approach carrying capacity
Immune killing	$\eta\, s\, b$	Immune cells destroy sensitive bacteria at rate η proportional to both populations
Antibiotic effect	$s\, \Sigma(\alpha_i + d_i)\, a_i$	Antibiotics kill sensitive bacteria at rate d_i and convert them to resistant through mutation at rate α_i

Differential Equation for Resistant Bacteria

$$\frac{dr}{dt} = \beta_R r(1 - s - r) - \eta r b + s[\alpha_1\, a_1 \;+\; \alpha_2\, a_2] \qquad \textbf{(2.2)}$$

The following table describes each term in the differential equation governing Resistant Bacteria:

Table 2.4: Breakdown of the Differential Equation for Resistant Bacteria

Term	Expression	Meaning
Logistic growth	$\beta_R r(1 - s - r)$	Resistant bacteria grow at rate β_R which is lower than β_S due to fitness cost c
Immune killing	$\eta r b$	Immune cells kill resistant bacteria at the same rate η as sensitive bacteria

Mutation gain	$s\Sigma\alpha_i\, a_i$	Sensitive bacteria mutate into resistant ones upon antibiotic exposure.

Differential Equation for Immune Cells

$$\frac{db}{dt} = kb\left(1 - \frac{b}{s+r}\right) \quad \textbf{(2.3)}$$

The following table describes each term in the differential equation governing Immune Cells:

Table 2.5: Breakdown of the Differential Equation for Immune Cells

Term	Expression	Meaning
Logistic growth	$kb\left(1 - \frac{b}{s+r}\right)$	Immune cells proliferate at rate k with carrying capacity equal to total bacterial load s+r. When bacteria decrease the immune response also contracts. When s+r = 0 immune cells decay at rate k

Differential Equation for Antibiotic Concentrations

$$\frac{da_1}{dt} = \mu_1(1 - a_1) \quad \textbf{(2.4)}$$

$$\frac{da_2}{dt} = \mu_2(1 - a_2) \quad \textbf{(2.5)}$$

The following table describes each term in the differential equation governing Antibiotic Concentrations:

Table 2.6: Breakdown of the Differential Equation for Antibiotic Concentrations

Term	Expression	Meaning
INH dynamics	$\mu_1(1 - a_1)$	INH is supplied continuously and cleared at rate μ_1 driving concentration toward normalized steady state of 1
PZA dynamics	$\mu_2(1 - a_2)$	Same structure as INH with its own clearance rate μ_2

Complete System

$$\frac{ds}{dt} = \beta_S\, s\,(1 - s - r) - \eta sb - s\,[(\alpha_1 + d_1)a_1 + (\alpha_2 + d_2)a_2]$$

$$\frac{dr}{dt} = \beta_R r(1 - s - r) - \eta rb + s\,[\alpha_1 a_1 + \alpha_2 a_2]$$

$$\frac{db}{dt} = kb\left(1 - \frac{b}{s+r}\right) \qquad \textbf{(2.6)}$$

$$\frac{da_1}{dt} = \mu_1(1 - a_1)$$

$$\frac{da_2}{dt} = \mu_2(1 - a_2)$$

Biological Constraints

The following table lists the biological constraints that must hold at all times for the system to remain valid:

Table 2.7: Biological Constraints and Their Interpretations

Constraint	Meaning
$s(t) \geq 0$	Sensitive bacteria cannot be negative
$r(t) \geq 0$	Resistant bacteria cannot be negative
$0 \leq b(t) \leq s+r$	Immune cells are bounded by bacterial load
$0 \leq a_i(t) \leq 1$	Antibiotic concentrations are normalized
$s(t) + r(t) \leq 1$	Total bacteria cannot exceed carrying capacity

2.2.4 Normalization of Variables

The original system before normalization is:

$$\frac{dS}{dt} = \beta_S S\left(1 - \frac{S+R}{T}\right) - \lambda SB - S\Sigma\bar{\alpha}_i A_i - S\Sigma\bar{d}_i A_i$$

$$\frac{dR}{dt} = \beta_R R\left(1 - \frac{S+R}{T}\right) - \lambda RB + S\Sigma\bar{\alpha}_i A_i \qquad \textbf{(2.7)}$$

$$\frac{dB}{dt} = kB\left(1 - \frac{B}{\omega(S+R)}\right)$$

$$\frac{dA_i}{dt} = \delta_i - \mu_i A_i \quad \text{for i = 1, 2}$$

Substitution Step

We introduce the following normalized variables:

$$s(t) = \frac{S(t)}{T} \qquad r(t) = \frac{R(t)}{T} \qquad b(t) = \frac{B(t)}{\omega T} \qquad a_i(t) = \frac{A_i(t)}{\frac{\delta_i}{\mu_I}}$$

With the new variables, the normalized system is given as:

$$\frac{ds}{dt} = \beta_S\, s\,(1 - s - r) - \eta sb - s\,[(\alpha_1 + d_1)a_1 \; + \; (\alpha_2 + d_2)a_2]$$

$$\frac{dr}{dt} = \beta_R r(1 - s - r) - \eta rb + s\,[\alpha_1 a_1 \; + \; \alpha_2 a_2]$$

$$\frac{db}{dt} = kb\left(1 - \frac{b}{s+r}\right) \tag{2.8}$$

$$\frac{da_1}{dt} = \mu_1(1 - a_1)$$

$$\frac{da_2}{dt} = \mu_2(1 - a_2)$$

The following table presents the reduced set of independent parameters obtained after the normalization step, where several original parameters are combined into new composite quantities:

Table 2.8: Reduced Independent Parameters After Normalization

New Parameter	Formula	Meaning
η	$\lambda\omega T$	Combined immune killing rate in normalized system
α_i	$\bar{\alpha}_i.(\delta_i/\mu_i)$	Normalized mutation rate for i^{th} antibiotic
d_i	$\bar{d}_i.(\delta_i/\mu_i)$	Normalized killing rate for i^{th} antibiotic
β_R	$(1 - c)\ \beta_S$	Resistant growth rate incorporating fitness cost

2.2.5 Threshold Parameters A, B and C

Three threshold parameters are derived from the model parameters. These parameters determine which equilibrium the system converges to and therefore determine the long-term outcome of the infection.

The following table defines the three threshold parameters used to determine the stability and outcome of the system:

Table 2.9: Definition and Interpretation of Threshold Parameters

Parameter	Formula	Meaning
A	$\frac{\beta_S - \Sigma\,(\alpha_i + d_i)}{\beta_S + \eta}$	Net Survival Ability of Sensitive Bacteria
B	$\frac{\beta_R}{\beta_R + \eta}$	Net Survival Ability of Resistant Bacteria
C	$\frac{\Sigma\alpha_i}{\beta_R + \eta}$	Mutation Contribution

Written explicitly for two antibiotics:

$$A = \frac{\beta_S - (\alpha_1 + d_1) - (\alpha_2 + d_2)}{\beta_S + \eta}$$

$$B = \frac{\beta_R}{\beta_R + \eta}$$

$$C = \frac{\alpha_1 + \alpha_2}{\beta_R + \eta}$$

The Key Threshold Condition

The following table summarizes the two threshold conditions and their corresponding biological outcomes at equilibrium:

Table 2.10: Threshold Conditions and Their Biological Outcomes

Condition	Outcome
$A < B$	Sensitive bacteria die out. Only resistant bacteria and immune cells persist at equilibrium E_2. Antibiotics have failed to prevent resistance dominance.
$A > B$	Both sensitive and resistant bacteria coexist with immune cells at equilibrium E_3. Neither type is eliminated.

$A < B \rightarrow E_2$ is stable → resistant bacteria dominate

$A > B \rightarrow E_3$ is stable → coexistence of both types

2.3 Mathematical Analysis

2.3.1 Equilibrium Points

Equilibrium points are found by setting all derivatives equal to zero and solving the resulting algebraic system.

$$\frac{ds}{dt} = 0$$

$$\frac{dr}{dt} = 0$$

$$\frac{db}{dt} = 0$$

$\frac{da_i}{dt} = 0$ for i = 1, 2

From $\frac{da_i}{dt} = 0$

⇨ $\mu_i\,(1 - a_i) = 0$

⇨ $a_i = 1$ *for i = 1, 2*

All equilibrium points have $a_1 = a_2 = 1$. Substituting $a_i = 1$ into the remaining three equations reduces the system to:

$\beta_S\, s\,(1 - s - r) - \eta\, s\, b - s\,.\Sigma(\alpha_i + d_i) = 0$ ... (i)

$\beta_R\, r\,(1 - s - r) - \eta\, r\, b\; +\; s \cdot \Sigma\alpha_i = 0$ ... (ii)

$k\, b\,(1\; - \frac{b}{s+r}) = 0$ ... (iii)

From equation (iii) either:

$b = 0$ (no immune response) or $b = s + r$ (immune cells at maximum capacity)

Case 1: When *b* = 0

Substituting b = 0 into (i) and (ii):

$$s\,[\beta_S\,(1 - s - r) - \Sigma(\alpha_i + d_i)] \;=\; 0$$

$$r\;\;[\beta_R \cdot (1 - s - r)] = 0$$

This gives s = 0 and r = 0

Subcase s = 0, r = 0 → gives E_0

Subcase s = 0, $r \neq$ 0 → from $\beta_R \cdot (1-r) = 0$ → r = 1 → gives E_1

Subcase $s \neq$ 0, $r \neq$ 0 → gives a point where s and r have opposite signs which is biologically meaningless. so, its rejected.

Case 2: When $b = s + r$

Substituting b = s+r into (i) and (ii):

$$s\,[\beta_S\,(1-s-r)-\eta\,(s+r)\;-\;\Sigma(\alpha_i+d_i)]\;=\;0 \qquad \text{... (iv)}$$

$$r\,[\beta_R\cdot(1-s-r)-\eta\,(s+r)]+\;s\Sigma\alpha_i\;=\;0 \qquad \text{... (v)}$$

Subcase s = 0 → substituting into (v):

- ⇨ $r\,[\beta_R\,(1-r)-\eta\,r]=0$
- ⇨ $r\,[\beta_R\;-\;(\beta_R+\eta)\,r]\;=\;0$
- ⇨ $r\;=\frac{\beta_R}{\beta_R+\eta}\;=\;B$ → gives E_2

Subcase $s \neq$ 0 → from (iv) the bracket equals zero:

- ⇨ $\beta_S\,(1-s-r)-\eta\,(s+r)-\Sigma(\alpha_i+d_i)=0$
- ⇨ $(\beta_S+\eta)\,(s+r)=\beta_S-\Sigma(\alpha_i+d_i)$
- ⇨ $(s+r)\;=\frac{\beta_S\,-\,\Sigma(\alpha_i+d_i)}{\beta_S+\eta}\;=A\;=\;\bar{b}$

Substituting back gives $\bar{s}$ and $\bar{r}$ → gives E_3

The following table summarizes the four equilibrium points of the system, including their values, existence conditions, and biological interpretations:

Table 2.11: Analysis of the four Equilibrium Points of the System Equations

Equilibrium	Values	Exists When	Biological Meaning
E_0	(0, 0, 0, 1, 1)	Always	System can always be infection free in theory
E_1	(0, 1, 0, 1, 1)	Always	Full resistant bacteria dominance always possible

E_2	$(0, B, B, 1, 1)$	Always	Resistant bacteria and immune coexistence always possible
E_3	$(\bar{s}, \bar{r}, \bar{b}, 1, 1)$	$A > B$ only	Coexistence only when sensitive bacteria are relatively strong

2.3.2 Stability Analysis

The Jacobian Matrix

Stability is determined by linearizing the system around each equilibrium point. This gives the Jacobian matrix J evaluated at each equilibrium:

$$J = \begin{bmatrix} \frac{\partial\varphi_1}{\partial s} & \frac{\partial\varphi_1}{\partial r} & \frac{\partial\varphi_1}{\partial b} & \frac{\partial\varphi_1}{\partial a_1} & \frac{\partial\varphi_1}{\partial a_2} \\ \frac{\partial\varphi_2}{\partial s} & \frac{\partial\varphi_2}{\partial r} & \frac{\partial\varphi_2}{\partial b} & \frac{\partial\varphi_2}{\partial a_1} & \frac{\partial\varphi_2}{\partial a_2} \\ \frac{\partial\varphi_3}{\partial s} & \frac{\partial\varphi_3}{\partial r} & \frac{\partial\varphi_3}{\partial b} & \frac{\partial\varphi_3}{\partial a_1} & \frac{\partial\varphi_3}{\partial a_2} \\ \frac{\partial\varphi_4}{\partial s} & \frac{\partial\varphi_4}{\partial r} & \frac{\partial\varphi_4}{\partial b} & \frac{\partial\varphi_4}{\partial a_1} & \frac{\partial\varphi_4}{\partial a_2} \\ \frac{\partial\varphi_5}{\partial s} & \frac{\partial\varphi_5}{\partial r} & \frac{\partial\varphi_5}{\partial b} & \frac{\partial\varphi_5}{\partial a_1} & \frac{\partial\varphi_5}{\partial a_2} \end{bmatrix}$$

Where $\varphi_1 \ldots \varphi_5$ are the right-hand sides of the five ODEs. Evaluated explicitly:

J=

$$\begin{bmatrix} \beta_s - 2\beta_s\,.s - \beta_s.r - \eta b - \Sigma(\alpha_i + d_i)a_I & -\beta_s.s & -\eta s & -s(\alpha_1 + d_1) & -s(\alpha_2 + d_2) \\ -\beta_R r + \Sigma\alpha_i a_i & \beta_R - \beta_R.s - 2\beta_R.r - \eta b & -\eta r & s\alpha_1 & s\alpha_2 \\ \frac{kb^2}{(s+r)^2} & \frac{kb^2}{(s+r)^2} & k - \frac{2kb}{s+r} & 0 & 0 \\ 0 & 0 & -\mu_1 & -\mu_1 & 0 \\ 0 & 0 & 0 & 0 & -\mu_2 \end{bmatrix}$$

The eigenvalues λ of J at each equilibrium determine stability:

All eigenvalues have negative real parts → equilibrium is Stable

Any eigenvalue has positive real part → equilibrium is unstable

Routh-Hurwitz Criteria

For a 5×5 Jacobian, finding eigenvalues directly requires solving a degree-5 polynomial which has no general closed form solution. The Routh-Hurwitz criteria allow us to determine whether all roots have negative real parts without solving for the roots explicitly.

The Criteria

Given characteristic polynomial:

$$P(\lambda) = \lambda^{n} + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \ldots + a_n = 0$$

All roots have negative real parts if and only if all Hurwitz determinants are positive.

For n = 2:

Conditions: $a_1 > 0$ and $a_2 > 0$

For n = 3:

Conditions: $a_1 > 0$ and $a_3 > 0$ and $a_1 \cdot a_2 > a_3$

Stability of E_0

$\lambda_1 = \beta_S - \Sigma(\alpha_i + d_i)$ (can be positive)

$\lambda 2 = \beta_R > 0$ (always positive)

Since $\lambda_2 = \beta_R$ is always positive by definition of parameters, E_0 is always unstable. The infection free state cannot be maintained.

Stability of E_1

$\lambda_1 = -\Sigma(\alpha_i + d_i) < 0$

$\lambda_2 = -\beta_R < 0$

$\lambda_3 = k > 0$ (always positive)

Since $\lambda_3 = k$ is always positive, E_1 is always unstable. Pure resistant bacteria without any immune response cannot be maintained.

Stability of E_2

Substituting E_2 = (0, B, B, 1, 1) into J

The eigenvalues from the last two rows are immediately:

$\lambda_4 = -\mu_1 < 0$

$\lambda_5 = -\mu_2 < 0$

These are always stable. The remaining three eigenvalues come from the block matrix:

$$JB(E_2) = \begin{bmatrix} \beta_S(1-B) - \eta \cdot B - \Sigma(\alpha_i + d_i) & 0 & 0 \\ -\beta_R \cdot B + \Sigma\alpha_I & -\beta_R \cdot B - \eta \cdot B & 0 \\ k & k & -k \end{bmatrix}$$

The characteristic equation factors as:

$(\beta_S(1-B) - \eta B - \Sigma(\alpha_i + d_i) - \lambda) \cdot (\lambda^2 + \lambda\,(k + \beta_R\, B) + k\,\beta_R\, B) = 0$

This gives:

$\lambda_{2,1} = \beta_S(1-B) - \eta B - \Sigma(\alpha_i + d_i)$

$= (\beta_S + \eta)\,(A - B)$

And $\lambda_{2,2}$ and $\lambda_{2,3}$ from:

$\lambda^2 + (k + \beta_R\, B)\,\lambda + k\,\beta_R\, B = 0$

Applying Routh-Hurwitz with n=2:

$a_1 = k + \beta_R\, B > 0$ (all parameters positive)

$a_2 = k\,\beta_R\, B > 0$ (all parameters positive)

So $\lambda_{2,2}$ and $\lambda_{2,3}$ always have negative real parts.

The only condition remaining is the sign of $\lambda_{2,1}$:

$\lambda_{2,1} = (\beta_S + \eta)\,(A - B)$

$A < B \rightarrow A - B < 0 \rightarrow \lambda_{2,1} < 0 \rightarrow$ E_2 is stable

$A > B \rightarrow A - B > 0 \rightarrow \lambda_{2,1} > 0 \rightarrow$ E_2 is unstable

Therefore, E_2 is stable if and only if $A < B$

Stability of E_3

Substituting $E_3 = (\bar{s}, \bar{r}, \bar{b}, 1, 1)$ into J

Again, the last two eigenvalues are:

$\lambda_4 = -\mu_1 < 0$

$\lambda_5 = -\mu_2 < 0$

The remaining three eigenvalues come from the block matrix:

$$JB(E_3) = \begin{bmatrix} -\beta_S \cdot \bar{s} & -\beta s \cdot \bar{s} & -\eta \cdot \bar{s} \\ -\beta_R \cdot \bar{r} + \Sigma\alpha_I & \beta_R\left(1-\bar{b}\right) - \eta \cdot \bar{b} - \beta_R \cdot \bar{r} & -\eta \cdot \bar{r} \\ k & k & -k \end{bmatrix}$$

The characteristic equation is:

$\lambda^3 + a_1\lambda^2 + a_2\lambda + a_3 = 0$

Where:

$a_1 = \bar{s}\,\bar{r}\,\Sigma\alpha_i + \beta_R\,\bar{r} + \beta_S\,\bar{s} + k$

$a_2 = k\,(\bar{s}\,\bar{r}\,\Sigma\alpha_i + \beta_R\,\bar{r} + \beta_S\,\bar{s}) + \bar{b}\,\eta\,k + \beta_S\,\bar{b}\,\bar{s}\,\bar{r}\,\Sigma\alpha_i$

$a_3 = k\,\eta\,\bar{b}\,\bar{s}\,\bar{r}\,\Sigma\alpha_i + k\,\beta_S\,\bar{b}\,\bar{s}\,\bar{r}\,\Sigma\alpha_i$

Applying Routh-Hurwitz with n=3:

$a_1 > 0$ (all terms positive since $\bar{s}, \bar{r}, \bar{b} > 0$ when $A > B$)

$a_3 > 0$ (all terms positive)

$a_1 \cdot a_2 > a_3$

All three Routh-Hurwitz conditions are satisfied whenever E_3 exists, that is whenever $A > B$.

Therefore, E_3 is stable if and only if $A > B$.

The following table summarizes the stability conditions for each of the four equilibrium points based on eigenvalue analysis:

Table 2.12: Stability Conditions for the Four Equilibrium Points

Equilibrium	Eigenvalue Signs	Stability Condition
E_0	At least one positive	Always unstable
E_1	At least one positive	Always unstable
E_2	All negative when $A < B$	$A < B$
E_3	All negative when $A > B$	$A > B$

Biological Interpretation of Each Equilibrium

The following table presents the biological interpretation of each equilibrium point in the context of disease progression and treatment outcome:

Table 2.13: Biological Interpretation of each Equilibrium Points

Equilibrium	Biological Meaning
E_0	Infection free state. Theoretically possible but always unstable. Any small bacterial introduction will grow and push the system away from this point.
E_1	Resistant bacteria completely dominate with no immune response. Always unstable. Any immune activation will disturb this state.
E_2	Sensitive bacteria are eliminated by antibiotics. Only resistant bacteria remain, held in balance by the immune system. This is the clinical outcome where resistance has won.
E_3	Both sensitive and resistant bacteria coexist with active immune response. Neither type is fully eliminated. Antibiotics alone are insufficient to clear the infection.

2.4 Results and Discussion

2.4.1 Parameter Values Used

All parameter values used in the numerical simulations are listed below.

The following tables present the parameter values and initial conditions used in the numerical simulation:

Table 2.14: Biological Parameter Values Used in Simulation

Symbol	Description	Value	Unit
β_S	Growth rate of sensitive bacteria	0.8	day^{-1}
β_R	Growth rate of resistant bacteria	0.4	day^{-1}
c	Fitness cost of resistance	0.5	dimensionless
η	Immune killing rate	0.3	day^{-1}
k	Immune cell proliferation rate	0.6	day^{-1}
ω	Ratio of immune cell capacity to bacterial capacity	1	dimensionless
T	Carrying capacity of bacteria	10^9	cells

Table 2.15: Antibiotic Parameter Values for Case I (A < B)

Symbol	Description	Value	Unit
α_I	Mutation rate due to INH	0.02	day^{-1}
d_I	Killing rate of sensitive bacteria by INH	0.15	day^{-1}
μ_I	Clearance rate of INH	0.06	day^{-1}
α_2	Mutation rate due to PZA	0.06	day^{-1}
d_2	Killing rate of sensitive bacteria by PZA	0.35	day^{-1}
μ_2	Clearance rate of PZA	0.03	day^{-1}

Table 2.16: Antibiotic Parameter Values for Case II (A > B)

Symbol	Description	Value	Unit
β_R	Growth rate of resistant bacteria	0.1	day^{-1}
η	Immune killing rate	0.05	day^{-1}
α_1	Mutation rate due to INH	0.06	day^{-1}
d_1	Killing rate of sensitive bacteria by INH	0.30	day^{-1}
α_2	Mutation rate due to PZA	0.02	day^{-1}
d_2	Killing rate of sensitive bacteria by PZA	0.05	day^{-1}
μ_1	Clearance rate of INH	0.06	day^{-1}
μ_2	Clearance rate of PZA	0.03	day^{-1}

Table 2.17: Initial Conditions for Numerical Simulation

Variable	Description	Value	Justification
$s(0)$	Initial sensitive bacteria	0.80	Heavy bacterial load at start of treatment
$r(0)$	Initial resistant bacteria	0.05	Small resistant population already present
$b(0)$	Initial immune cells	0.05	Weak immune response at start
$a_1(0)$	Initial INH concentration	0.05	Antibiotic just administered
$a_2(0)$	Initial PZA concentration	0.02	Antibiotic just administered

2.4.2 Threshold Values A, B and C

The following table shows the computed values of A, B, C for the case I and II:

Table 2.18: Computed values of A, B, C for Case I and II

Parameter	Formula	Case I	Case II
A	$\frac{\beta_S - \Sigma(\alpha_i + d_i)}{\beta_S + \eta}$	0.2000	0.3364
B	$\frac{\beta_R}{\beta_R + \eta}$	0.5714	0.2500
C	$\frac{\Sigma \alpha_i}{\beta_R + \eta}$	0.1143	0.0267
Condition	—	$A < B$	$A > B$
Stable Equilibrium	—	E_2	E_3

In Case I where $A < B$, the antibiotics and immune system together are strong enough to eliminate sensitive bacteria completely. However resistant bacteria survive because they are unaffected by antibiotics and can only be controlled by the immune system. The system converges to E_2 where only resistant bacteria and immune cells coexist at equilibrium. This represents the clinically unfavorable outcome where antibiotic resistance has effectively won.

In Case II where $A > B$, the antibiotics are too weak relative to the growth advantage of sensitive bacteria, or the resistant bacteria have a very high fitness cost making them slow to grow. Neither bacterial type is fully eliminated and the system converges to E_3 where sensitive bacteria, resistant bacteria, and immune cells all coexist at a stable positive equilibrium.

2.4.3 Case I: *A* < *B*

All simulations in this section use the Case I parameters where A = 0.2000 and B = 0.5714, giving the condition A < B. The system is therefore expected to converge to equilibrium E_2 = (0, 0.5714, 0.5714, 1, 1).

The following figure shows the time evolution of all five state variables under Case I parameters where $A = 0.2000 < B = 0.5714$.

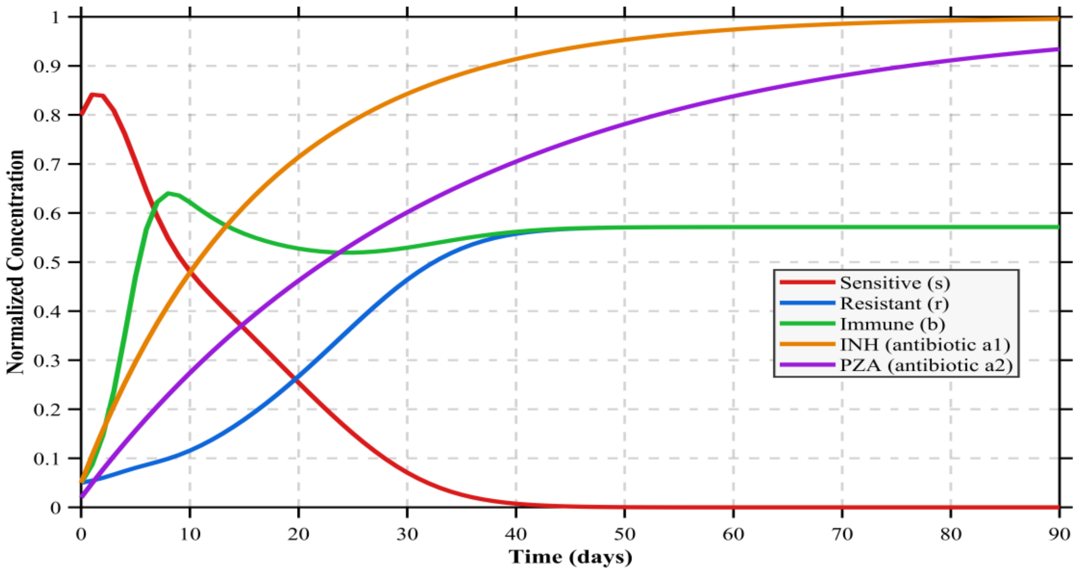

Fig: 2.1 All 5 Variables Over Time (*A*<*B*)

The following figure isolates the relationship between total bacterial load and immune cell population over time:

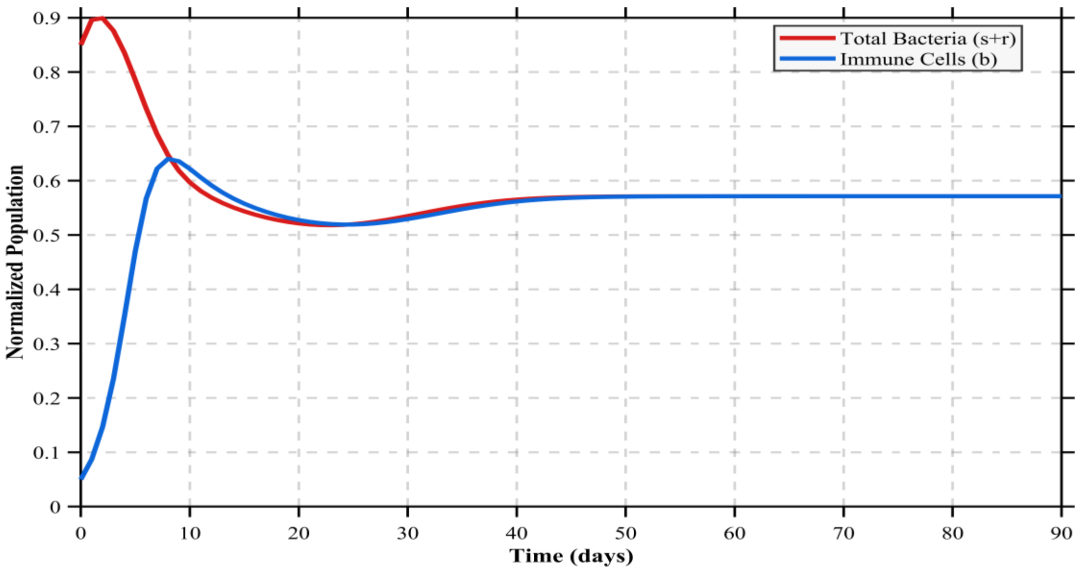

Fig: 2.2 Total Bacteria vs Immune Cells

The following figure shows the pharmacokinetic behavior of both antibiotics over the 200-day treatment period.

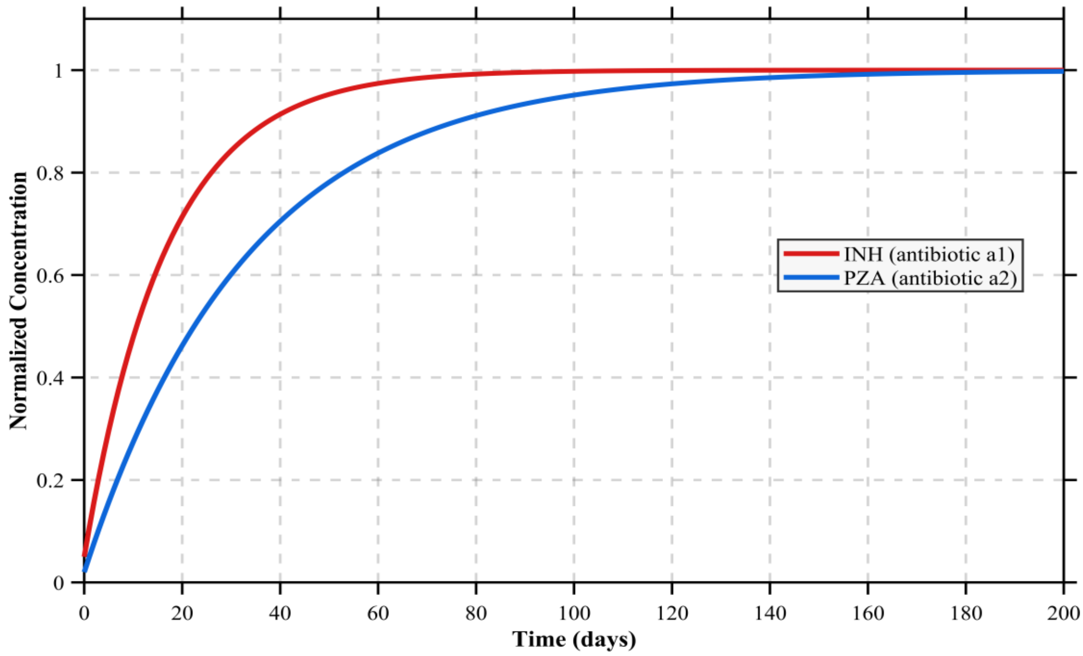

Fig: 2.3 Antibiotic Concentrations

The following phase Plot shows the trajectory of sensitive bacteria s against resistant bacteria r over the 90-day period under Case I conditions.

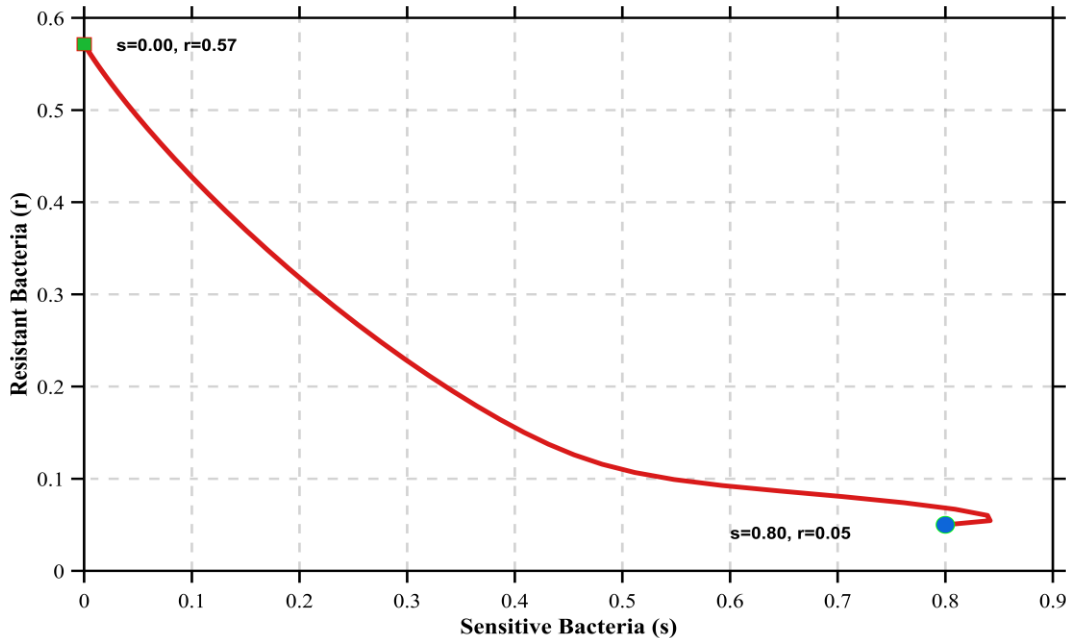

Fig: 2.4 Phase Portrait (s vs r)

The numerical simulations confirm the analytical predictions for Case I. Sensitive bacteria are eliminated by day 60 under the combined effect of antibiotics and the immune system, while resistant bacteria and immune cells stabilize at the same equilibrium level (Figure 2.1). Both antibiotics reach their maximum concentration within 30 days and remain constant thereafter, with INH stabilizing faster than PZA due to its higher clearance rate (Figure 2.3). The immune system tracks the total bacterial load throughout, with both curves converging to the same value by day 90, confirming a stable but not infection-free balance (Figure 2.2). The phase portrait confirms full convergence to E_2 at $s = 0$, $r = 0.5714$, with the trajectory terminating on the r-axis in contrast to Case II where the trajectory ends at a positive s value, confirming coexistence (Figure 2.4).

Numerical Results for Case I ($A < B$)

The following table records exact numerical values at 10-day intervals confirming convergence to $E_2 = (0, 0.5714, 0.5714, 1, 1)$.

Table: 2.1 Numerical Results for Case I (A < B)

Time (days)	Sensitive (s)	Resistant (r)	Total (s+r)	Immune Cells (b)	INH (a_1)	PZA (a_2)
0	0.8000	0.0500	0.8500	0.0500	0.0500	0.0200
10	0.4812	0.1157	0.5969	0.6219	0.4786	0.2740
20	0.2542	0.2671	0.5213	0.5275	0.7139	0.4622
30	0.0708	0.4641	0.5349	0.5294	0.8430	0.6016
40	0.0075	0.5577	0.5651	0.5617	0.9138	0.7048
50	0.0004	0.5706	0.5710	0.5707	0.9527	0.7813
60	0.0000	0.5714	0.5714	0.5714	0.9740	0.8380
70	0.0000	0.5714	0.5714	0.5714	0.9858	0.8800
80	0.0000	0.5714	0.5714	0.5714	0.9922	0.9111
90	0.0000	0.5714	0.5714	0.5714	0.9957	0.9341

Sensitive bacteria fall to zero by day 60 and remain there. Resistant bacteria and immune cells both stabilize at 0.5714 by day 90, numerically confirming the theoretical equilibrium E_2.

2.4.4 Case II: $A > B$

All simulations in this section use the Case II parameters where $A = 0.3364$ and $B = 0.2500$, giving the condition $A > B$. The system is therefore expected to converge to equilibrium $E_3 = (0.1015, 0.2350, 0.3364, 1, 1)$.

The following figure shows the time evolution of all five state variables under Case II parameters where $A = 0.3364 > B = 0.2500$.

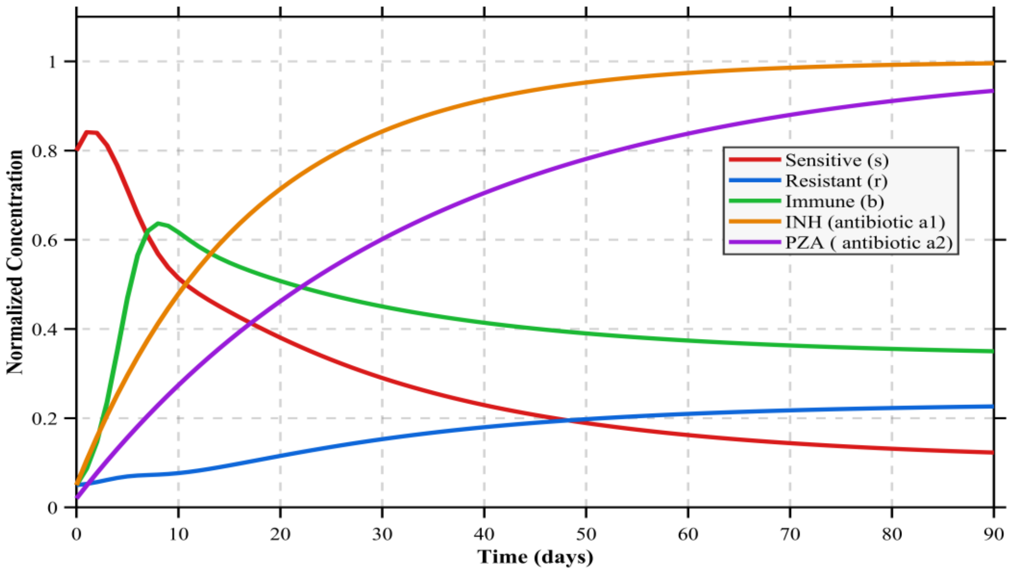

Fig: 2.5 All 5 Variables Over Time (A>B)

The following phase portrait shows the trajectory of sensitive bacteria s against resistant bacteria r over the 90-day period.

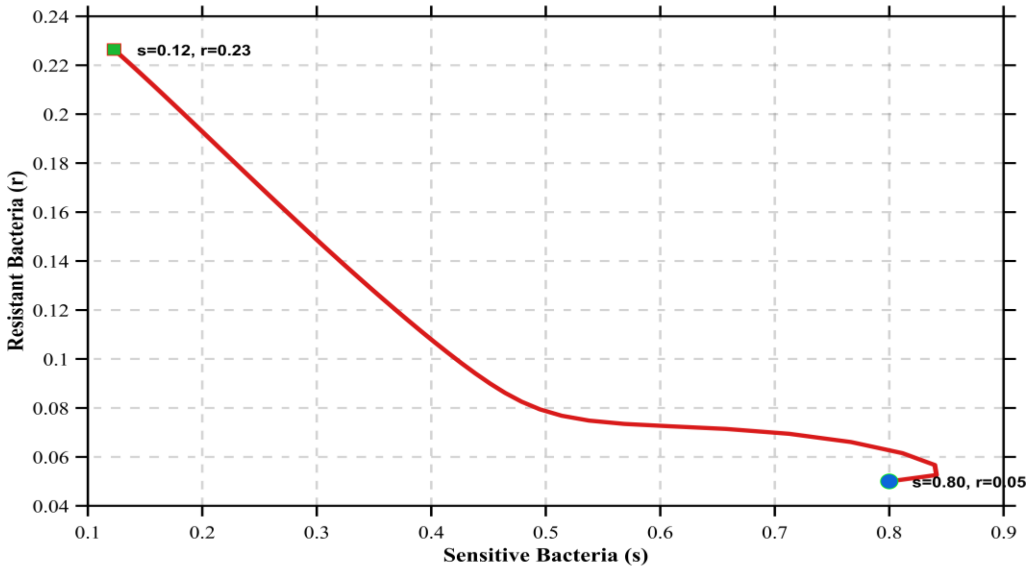

Fig: 2.6 Phase Portrait (s vs r)

The following figure shows all three biological populations together to demonstrate coexistence under Case II conditions.

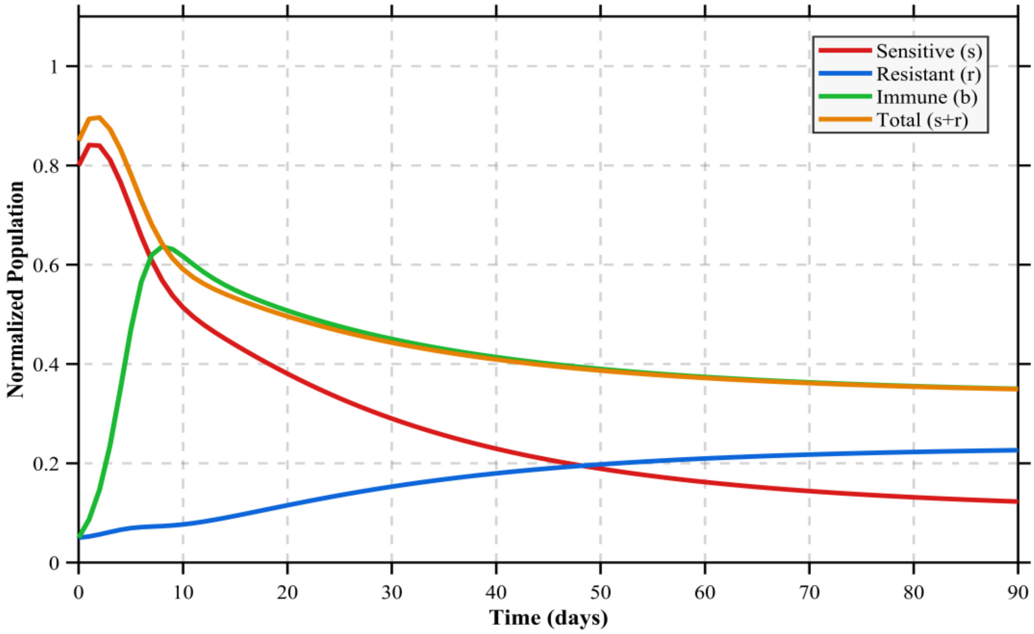

Fig: 2.7 Bacteria and Immune Coexistence

The numerical simulations confirm the analytical predictions for Case II. Unlike Case I, sensitive bacteria do not reach zero but instead decline and stabilize at a positive value, coexisting alongside resistant bacteria and immune cells at equilibrium E_3 (Figure 2.5). The phase portrait confirms this, with the trajectory moving from the initial condition toward E_3 where both s and r remain positive — visually confirming that neither bacterial strain is eliminated (Figure 2.6). All three populations stabilize at positive values by day 90, reflecting a three-way balance in which the combined effect of antibiotics and the immune system is insufficient to clear the infection (Figure 2.7). In contrast to Case I, where immune cells converged to the same value as resistant bacteria alone, in Case II the immune cells track the total bacterial load $s + r$, which remains higher due to the persistent coexistence of both bacterial strains.

Numerical Results for Case II ($A > B$)

The following table records exact numerical values at 20-day intervals showing convergence toward E_3.

Table: 2.2 Numerical Results for Case II ($A > B$)

Time (days)	Sensitive (s)	Resistant (r)	Total (s+r)	Immune Cells (b)	INH (a_1)	PZA (a_2)
0	0.8000	0.0500	0.8500	0.0500	0.0500	0.0200
20	0.3805	0.1155	0.4960	0.5075	0.7139	0.4622
40	0.2293	0.1798	0.4091	0.4139	0.9138	0.7048
60	0.1622	0.2096	0.3718	0.3740	0.9740	0.8380
80	0.1315	0.2228	0.3543	0.3554	0.9922	0.9111
100	0.1169	0.2288	0.3457	0.3462	0.9976	0.9512
120	0.1095	0.2318	0.3413	0.3416	0.9993	0.9732
140	0.1058	0.2333	0.3390	0.3392	0.9998	0.9853
160	0.1038	0.2340	0.3378	0.3379	0.9999	0.9919
180	0.1027	0.2344	0.3371	0.3372	1.0000	0.9956
200	0.1021	0.2347	0.3368	0.3368	1.0000	0.9976
220	0.1018	0.2348	0.3366	0.3366	1.0000	0.9987
240	0.1016	0.2348	0.3365	0.3365	1.0000	0.9993

The table records numerical values at 20-day intervals up to day 240. By day 220 the system has effectively converged with changes between consecutive time points less than 0.0002. All three populations s, r and b remain positive throughout. s stabilizes at 0.1016, r at 0.2348, and b at 0.3365 which numerically confirming the theoretical equilibrium E_3 = (0.1015, 0.2350, 0.3364, 1, 1) Unlike Case I where sensitive bacteria reached zero, in Case II all populations coexist at positive values as predicted by the $A > B$ threshold condition.

2.4.5 Numerical Analysis of Model Parameters and Treatment Strategies

The following figure shows total bacterial load over 90 days for four values of η (0.1, 0.3, 0.6, 1.0), where η represents the rate at which immune cells kill bacteria.

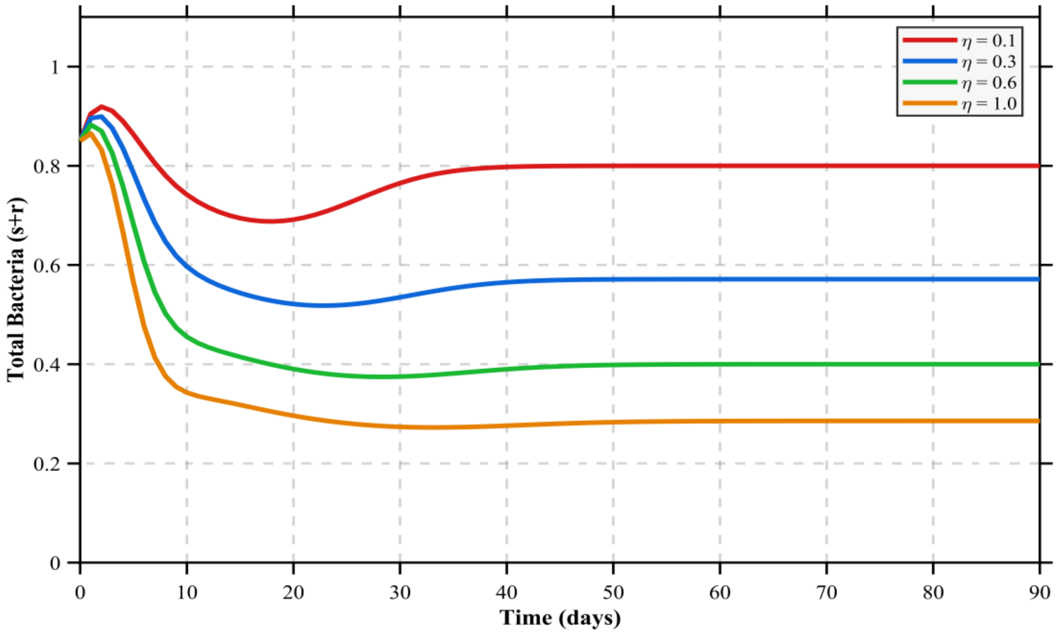

Fig: 2.8 Effect of Immune Strength on Total Bacterial load

The following figure shows how the resistant bacterial population evolves over 90 days for four values of fitness cost c (0.1, 0.3, 0.5, 0.9), where c directly reduces the growth rate of resistant bacteria as $\beta_R = (1-c)\, \beta_S$.

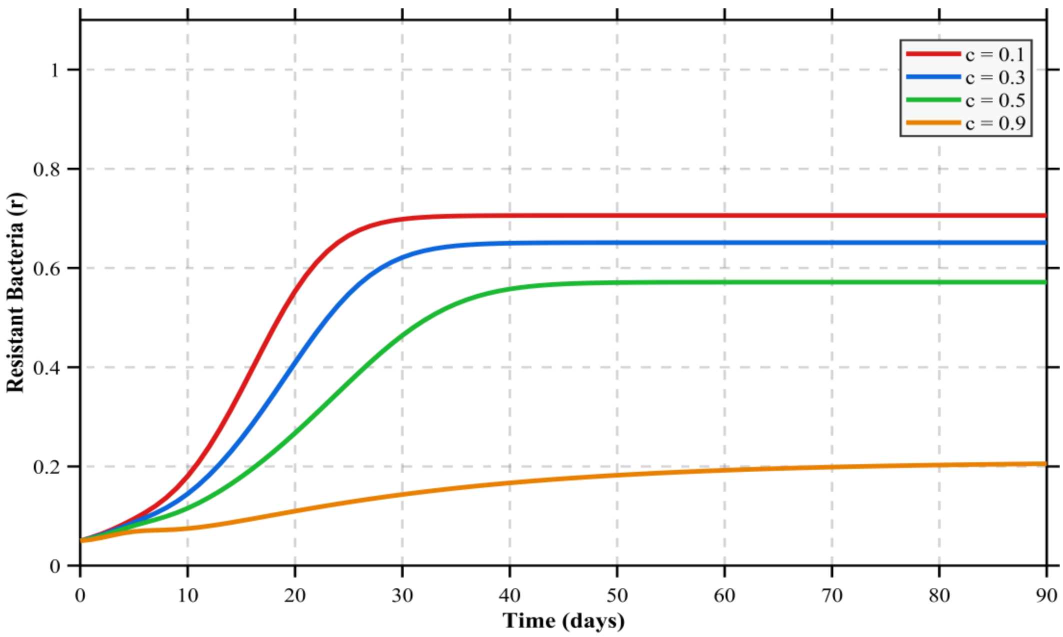

Fig: 2.9 Effect of Fitness Cost in Resistance Bacteria

The plot compares total bacterial load when antibiotics are stopped at day 30, day 60, or completed for the full 90-day course.

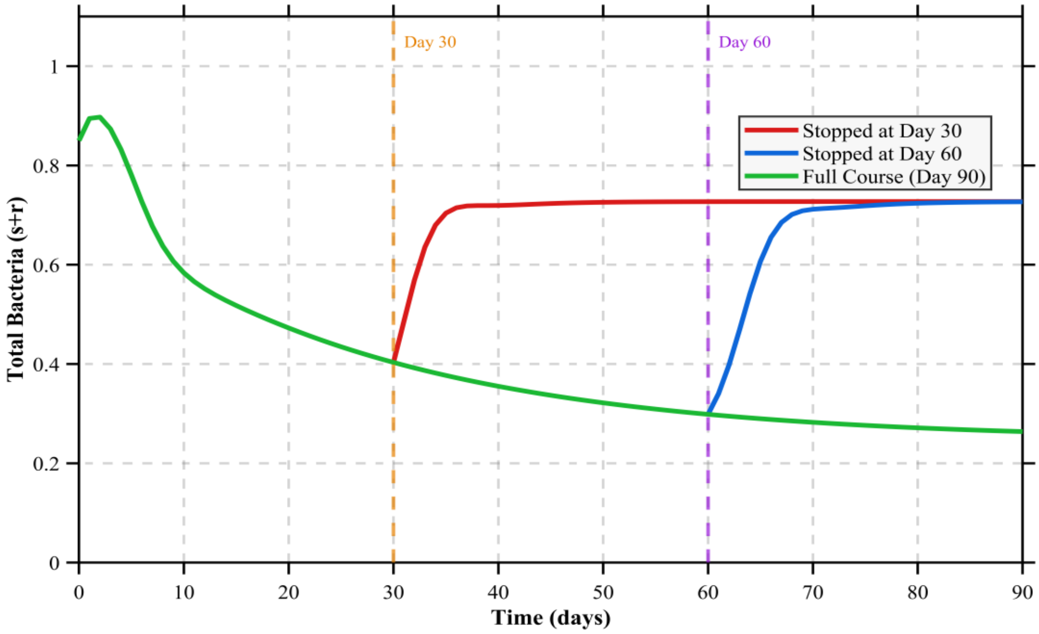

Fig: 2.10 Effect of Antibiotic Termination on Total Bacterial load

The following plot shows sensitive bacteria *s* only, which more clearly demonstrates the difference in clearance speed between the three treatment strategies.

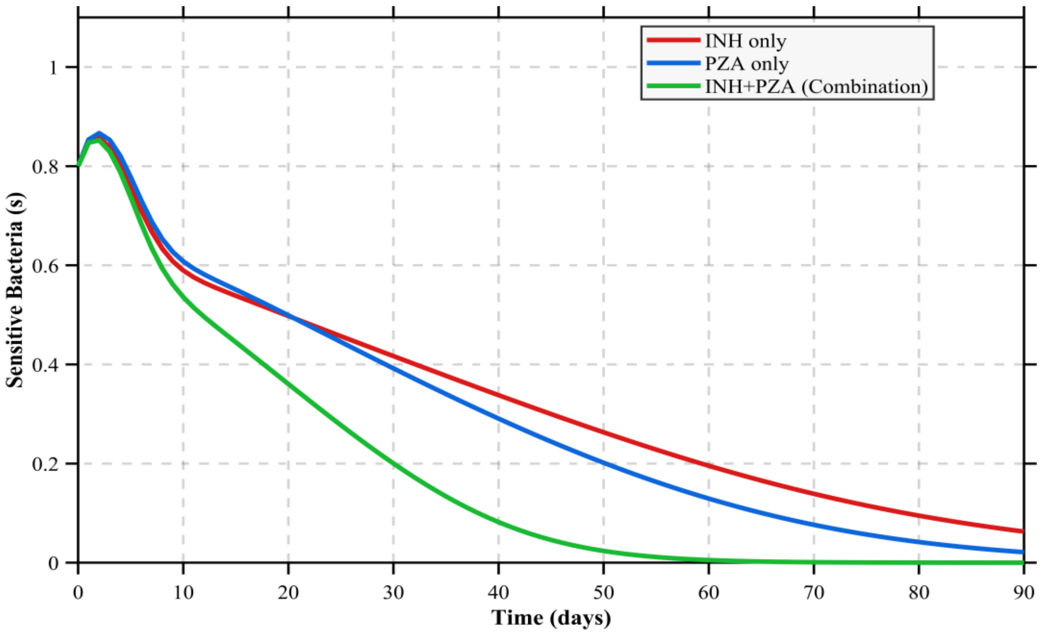

Fig: 2.11 Effect of Single vs Combination Therapy on sensitive bacteria

The following plot compares total bacterial load under three treatment strategies using identical initial conditions.

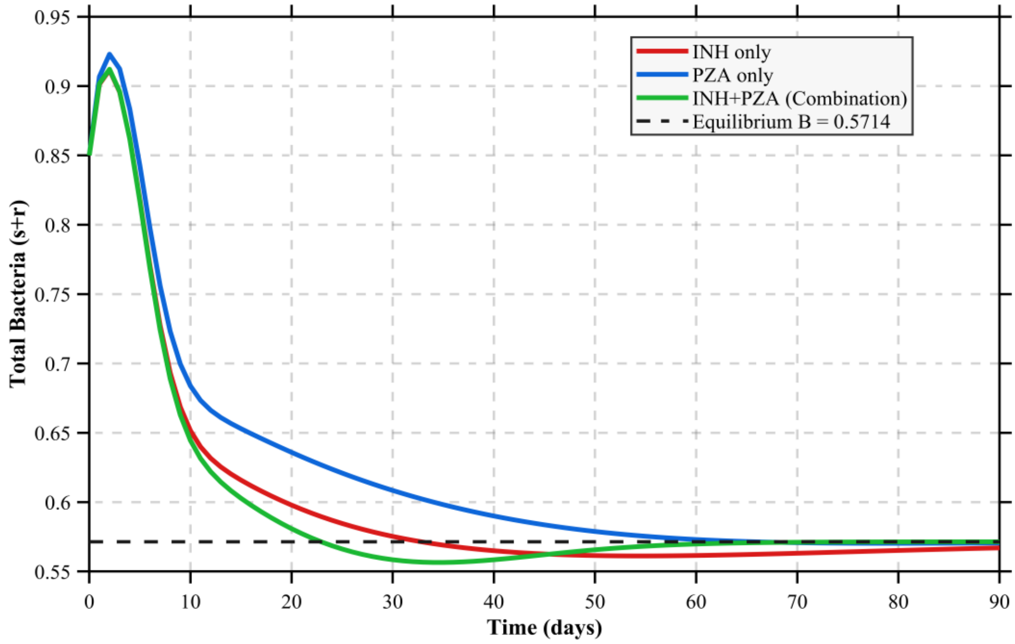

Fig: 2.12 Effect of Single vs Combination Therapy on Total bacteria

The simulations reveal four key clinical insights. A stronger immune response (higher η) directly lowers the equilibrium bacterial load, with η = 1.0 producing the most aggressive bacterial suppression confirming that immunocompromised patients carry a higher burden even under identical antibiotic treatment (Figure 2.8). The fitness cost of resistance acts as a natural limiting factor; as c increases, resistant bacteria grow more slowly and stabilize at lower levels, suggesting that high-cost resistant strains can die out naturally when antibiotic pressure is reduced (Figure 2.9). Early termination of antibiotics leads to bacterial rebound stopping at day 30 causes a sharp resurgence, stopping at day 60 causes a partial rebound, while completing the full 90-day course maintains the system at E_2 throughout, confirming that incomplete treatment actively creates conditions for bacterial resurgence (Figure 2.10). Finally, while all three treatment strategies INH alone, PZA alone, and combination therapy, ultimately converge to the same E_2 equilibrium at $s + r \approx 0.5714$, combination therapy achieves the fastest clearance of sensitive bacteria and the fastest convergence to equilibrium, offering the greatest suppression of bacterial load throughout the treatment period (Figures 2.11 and 2.12).

The following table records the total bacterial load (s+r) at 20-day intervals for all three treatment strategies, confirming convergence to the theoretical E_2 equilibrium at 0.5714.

Table 2.3: Total Bacterial Load (s+r) Under Three Treatment Strategies

Time (days)	INH only	PZA only	Combination
0	0.8500	0.8500	0.8500
20	0.5979	0.6359	0.5809
40	0.5650	0.5900	0.5585
60	0.5615	0.5731	0.5698
80	0.5652	0.5703	0.5714
100	0.5685	0.5709	0.5714
120	0.5702	0.5713	0.5714
140	0.5709	0.5714	0.5714
160	0.5712	0.5714	0.5714
180	0.5714	0.5714	0.5714

All three treatment strategies converge to the same E_2 equilibrium value of 0.5714 by day 180, confirming the theoretical prediction. Combination therapy converges fastest, reaching equilibrium by day 80. PZA alone reaches equilibrium by day 140, while INH alone is the slowest, stabilizing at 0.5714 only by day 180. This numerical result reinforces the conclusion that while all three strategies ultimately reach the same long-run equilibrium, combination therapy provides the fastest bacterial suppression throughout the treatment period.

Chapter 3

Logistic Population Models with Harvesting

3.1 Overview

3.1.1 Malthusian Growth Model

The simplest population model was proposed by Thomas Malthus in 1798. It assumes that the rate of population growth is directly proportional to the current population size:

$$\frac{dP}{dt} = kP \qquad \textbf{(3.1)}$$

where k > 0 is the intrinsic growth rate. The solution to this equation is:

$$P(t) = P_0\, e^{kt} \qquad \textbf{(3.2)}$$

This describes unlimited exponential growth. While realistic for small populations with abundant resources, it fails in practice because it predicts that the population grows without bound, which is biologically impossible. This fundamental flaw motivated the development of the logistic growth model.

3.1.2 Logistic Growth Model

To correct the unrealistic assumption of unlimited growth, Pierre François Verhulst introduced the logistic growth model in 1838 by incorporating a carrying capacity *N* i.e. the maximum population size the environment can sustainably support. The logistic equation is:

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) \qquad \textbf{(3.3)}$$

The term (1 - *P/N*) acts as a braking factor. When *P* is much smaller than *N*, the term is close to 1 and growth is nearly exponential. As *P* approaches *N*, the term approaches 0 and growth slows. If *P* exceeds *N*, the term becomes negative and the population declines back toward *N*. The analytic solution is:

$$P(t) = \frac{N}{1 + \left(\frac{N - P_0}{P_0}\right) e^{(-kt)}} \quad \textbf{(3.4)}$$

This produces the characteristic S-shaped (sigmoidal) growth curve observed in many real populations.

3.2 Logistic Growth with Constant Harvesting

3.2.1 Model Formulation

In the constant harvesting model, it is assumed that a fixed number of fish, denoted by a, are removed from the population per unit time regardless of the current population size. Adding this harvesting term to the logistic growth equation gives the following first-order ordinary differential equation:

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) - a \quad \textbf{(3.5)}$$

The parameters appearing in equation 3.5 are described in the following table:

Table 3.1: Parameters of the Constant Harvesting Model

Parameter	Description	Unit
P	Size of the fish population at time t	number of fish
t	Time	years or seasons
k	Intrinsic growth rate	per unit time
N	Carrying capacity of the environment	number of fish
a	Constant harvesting rate	fish per unit time

This equation states that the rate of change of the population at any time is equal to the natural logistic growth minus the constant amount removed by harvesting. When a = 0, the equation reduces to the standard logistic model with no harvesting.

3.2.2 Equilibrium Analysis

Equilibrium points are found by setting $\frac{\mathrm{dP}}{\mathrm{dt}} = 0$

⇨ $kP\,(1 - \frac{P}{N}) - a = 0$

Expanding the left-hand side:

⇨ $kP - \frac{kP^2}{N} - a = 0$

Multiplying through by -N/k:

⇨ $P^2 - NP + \frac{aN}{k} = 0$

This is a quadratic equation in P with coefficients:

a = 1, b = -N, c = $\frac{aN}{k}$

Applying the quadratic formula $P = \frac{-b \pm \sqrt{(b^2-4ac)}}{2a}$:

$$P = \frac{N \pm \sqrt{\left(N^2 - \frac{4aN}{k}\right)}}{2}$$

This gives two equilibrium points:

$$P_1 = \frac{N + \sqrt{\left(N^2 - \frac{4aN}{k}\right)}}{2}$$

$$P_2 = \frac{N - \sqrt{\left(N^2 - \frac{4aN}{k}\right)}}{2}$$

The existence of real equilibrium points depends entirely on the discriminant:

$$D = N^2 - \frac{4aN}{k}$$

3.2.3 Discriminant Condition

The discriminant $D = N^2 - \frac{4aN}{k}$ determines the number and nature of equilibrium points. Three cases arise:

Case 1: *D* > 0: Two distinct equilibria

Condition: $a < \frac{kN}{4}$

Two distinct real equilibrium points P_1 and P_2 exist. The population can survive if the initial condition is above P_2.

Case 2: *D* = 0: One equilibrium (MSY point)

Condition: $a = \frac{kN}{4}$

The two equilibria merge into a single point at $P^* = N/2$. This is the critical boundary called the Maximum Sustainable Yield. Any harvesting beyond this point leads to extinction.

Case 3: *D* < 0: No real equilibria

Condition: $a > \frac{kN}{4}$

No real equilibrium points exist. The harvesting rate exceeds what the population can recover from and the population declines to extinction regardless of the initial condition.

3.2.4 Maximum Sustainable Yield

The Maximum Sustainable Yield (MSY) is the largest harvesting rate a that still permits the population to sustain itself indefinitely. It is found by setting the discriminant equal to zero:

$D = 0$

$$N^2 - \frac{4aN}{k} = 0$$

Solving for a:

$$MSY = \frac{kN}{4}$$

For D=0:

$$P1, P2 = \frac{N + 0}{2} = \frac{N}{2}$$

At this harvesting rate the two equilibria merge at:

$$P^* = \frac{N}{2}$$

This means the MSY is achieved when the population is maintained at exactly half its carrying capacity. For the parameters (k = 0.20, N = 5):

MSY = (0.20 x 5) / 4 = 0.25

Any harvesting rate above 0.25 will drive the population to extinction.

3.2.5 Stability Analysis

To classify the stability of each equilibrium point, the linearization method is applied.

Let:

$$f(P) = kP\,(1 - \tfrac{P}{N}) - a \qquad \textbf{(3.6)}$$

The derivative with respect to P is:

$$\frac{df}{dP} = k\,(1 - \tfrac{2P}{N}) \qquad \textbf{(3.7)}$$

At P_1:

Since $P_1 = (N + \text{sqrt}(D))/2$ is greater than $N/2$, substituting into the derivative gives:

$$\frac{df}{dP} = \mathrm{k}\,(1 - \tfrac{2\mathrm{P}_1}{\mathrm{N}}) < 0$$

Since the eigenvalue is negative, P_1 is a stable equilibrium. Populations near P_1 will return to it after a small disturbance.

At P_2:

Since P_2 = (N - sqrt(D))/2 is less than N/2, substituting into the derivative gives:

$$\frac{df}{dP} = \mathrm{k}\left(1 - \tfrac{2\mathrm{P}_2}{\mathrm{N}}\right) > 0$$

Since the eigenvalue is positive, P_2 is an unstable equilibrium. Populations above P_2 recover toward P_1, while populations below P_2 decline to extinction. This result has a critical biological interpretation: P_2 acts as a danger threshold. If the fish population ever falls below P_2 due to overharvesting, environmental shock, or disease. It will continue to decline and cannot recover on its own.

3.2.6 Parameter Table and Equilibrium Values

Using the parameters $k = 0.20$ and $N = 5$, the equilibrium points are computed as:

$P_1, P_2 = (5 \pm \sqrt{25 - 20a}) / 2$

The following table explains the Equilibrium Points for Verifying the Harvesting Rate:

Table:3.2: Equilibrium Points for Varying Harvesting Rate ($k = 0.20$, $N = 5$)

a	Discriminant D	P_1 (stable)	P_2 (unstable)	Status
0.00	25.0000	5.0000	0.0000	Two equilibria
0.21	4.000	3.5000	1.5000	Two equilibria
0.22	3.000	3.3660	1.6340	Two equilibria
0.23	2.000	3.2071	1.7929	Two equilibria
0.24	1.000	3.0000	2.0000	Two equilibria
0.25	0.000	2.5000	2.5000	MSY
0.26	-1.000	—	—	Extinction

As a increases, P_1 decreases and P_2 increases the two equilibria move closer together. At a = 0.25 they meet at $P^* = 2.5$. Beyond this point no sustainable equilibrium exists.

3.2.7 Results and Discussion

The following figures present the graphical results of the constant harvesting model using the parameters k = 0.20, N = 5, and varying values of the harvesting rate a. Five figures are presented: the equilibrium map, the phase portrait, and three slope field plots corresponding to the three discriminant cases identified in Section 3.2.3.

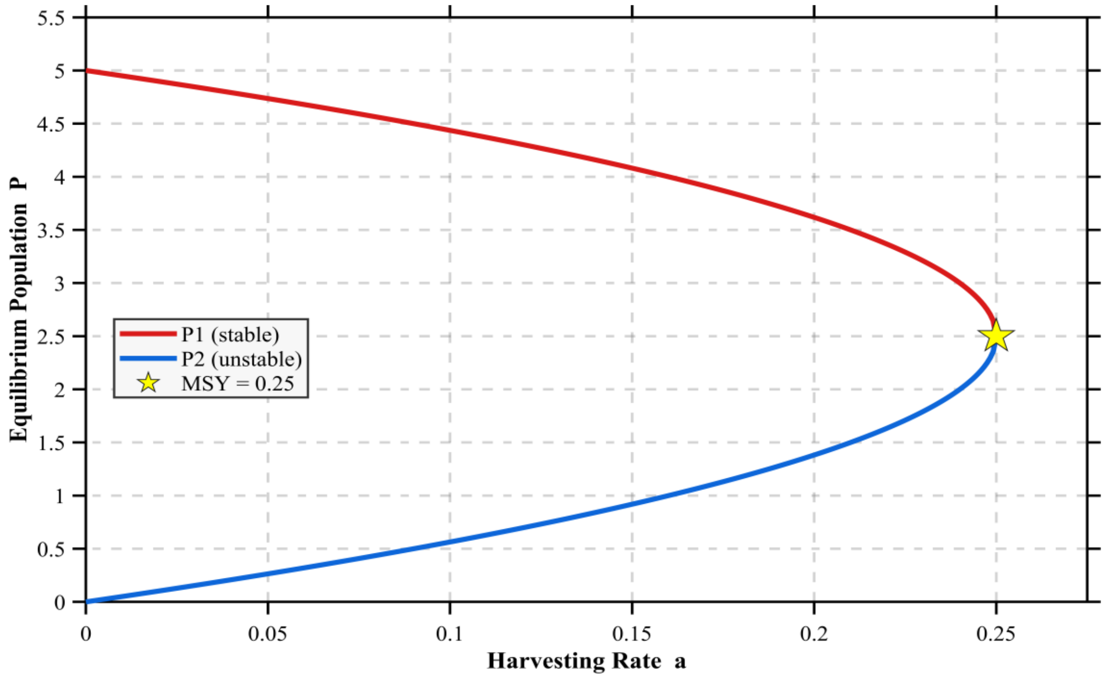

Fig 3.1: Equilibrium Map (P_1 and P_2 vs a)

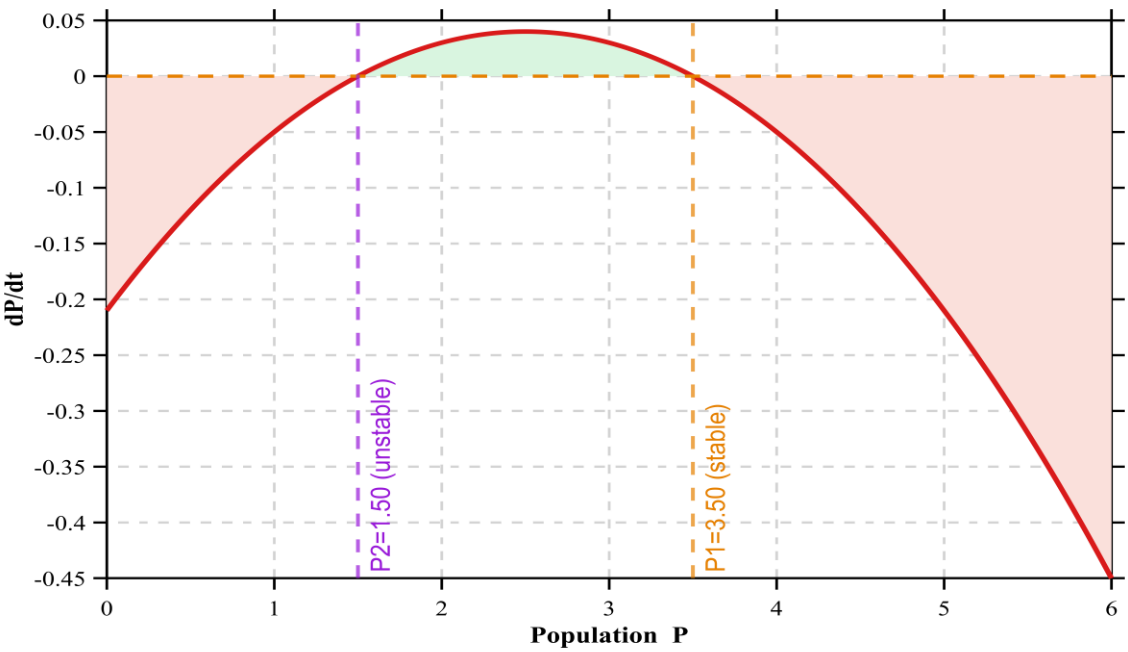

Fig 3.2: Phase Portrait (dP/dt vs P)

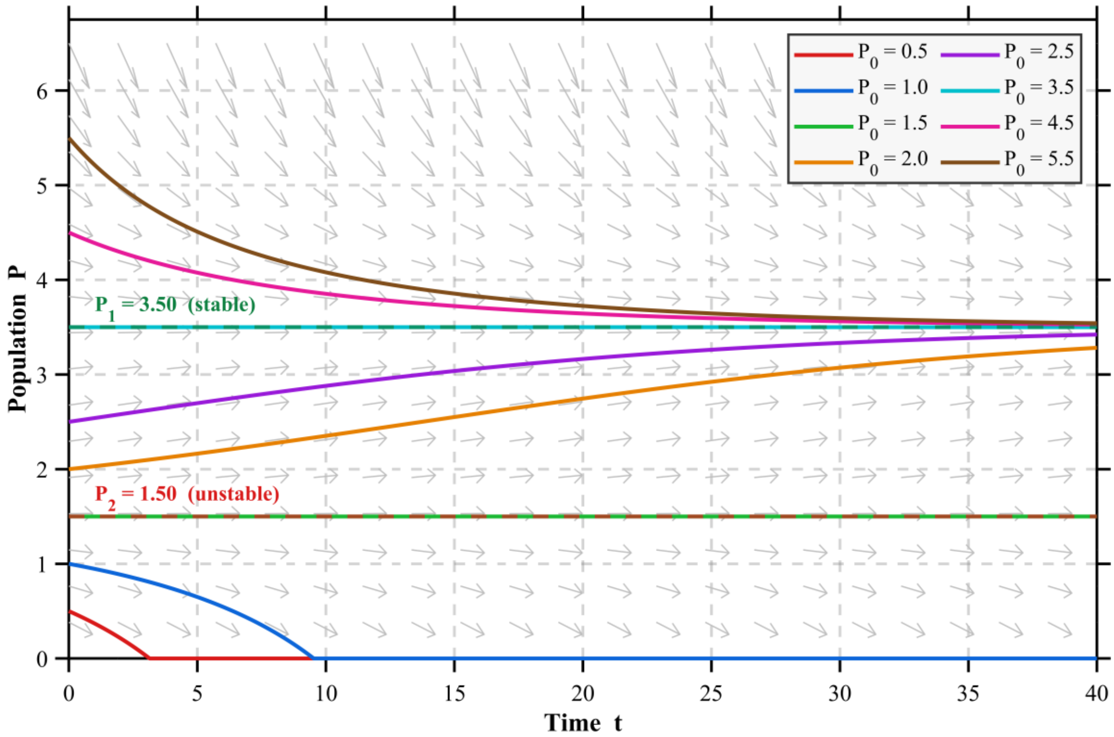

Fig 3.3: Slope Field and Trajectories for $a <$ MSY ($a = 0.21$)

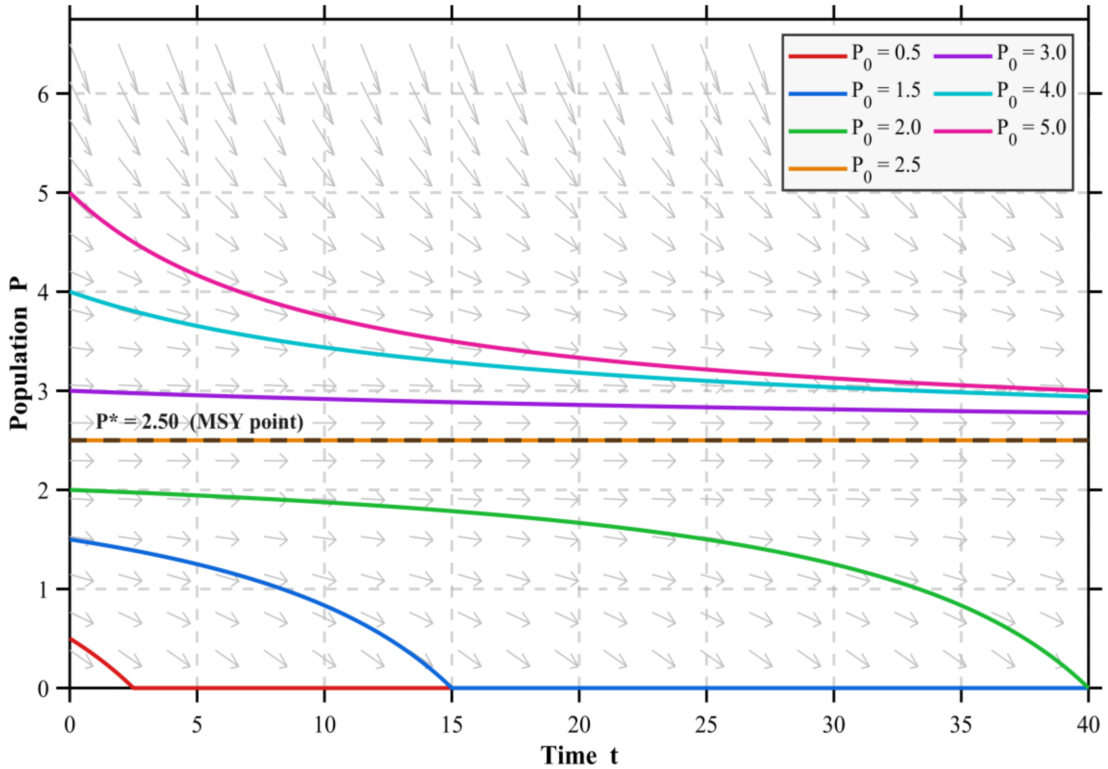

Fig 3.4: Slope Field and Trajectories at $a =$ MSY ($a = 0.25$)

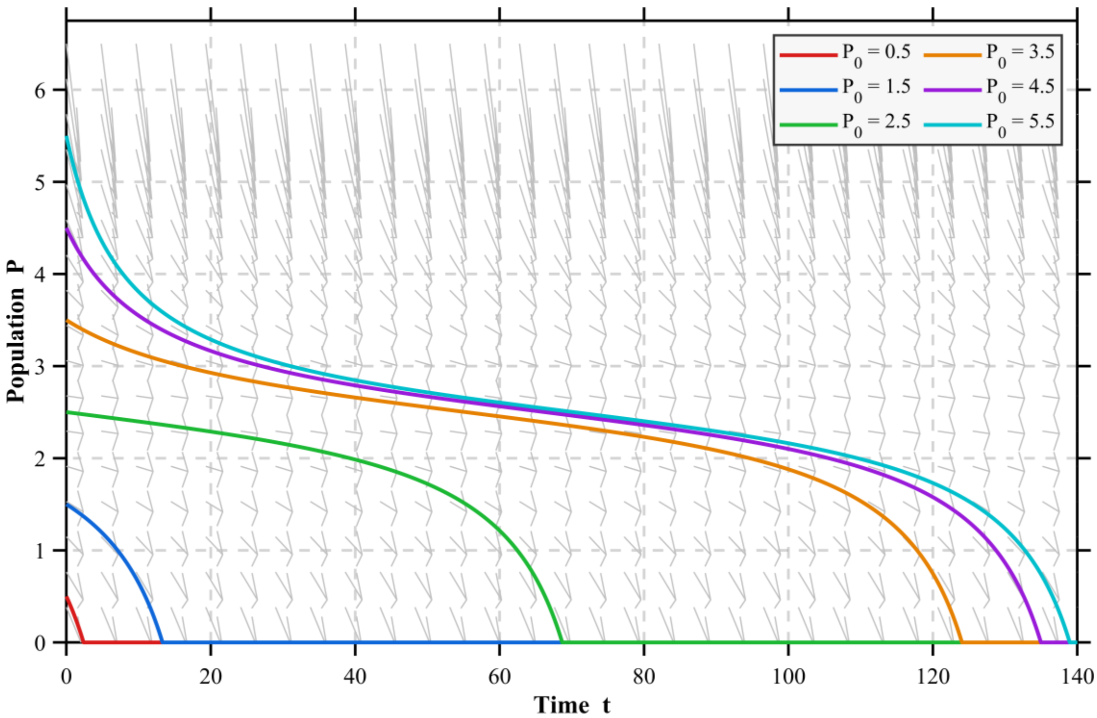

Fig 3.5: Slope Field and Trajectories for a > MSY (a = 0.26)

The simulations confirm the analytical results for the constant harvesting model across all three discriminant cases. The equilibrium map shows P_1 decreasing and P_2 increasing as the harvesting rate a rise, with both curves meeting at the MSY point at a = 0.25 and P = 2.5, forming a critical turning point beyond which no real equilibria exist and extinction becomes inevitable (Figure 3.1). The phase portrait for a = 0.21 confirms the two equilibria at P_2 = 1.50 (unstable) and P_1 = 3.50 (stable), with the growth zone between them and decline zones outside, consistent with the analytical zero-crossing analysis (Figure 3.2).

The slope field for a < MSY shows that all trajectories starting above P_2 = 1.5 converge to the stable equilibrium P_1 = 3.5, while those starting below P_2 decline monotonically to extinction, confirming the danger threshold behavior of the unstable equilibrium (Figure 3.3). At exactly a = MSY, trajectories above P^* = 2.5 converge toward it extremely slowly due to the marginally stable nature of the merged equilibrium where df/dP = 0, while those below still go extinct, placing the system at its most delicate balance (Figure 3.4). When a exceeds the MSY at a = 0.26, the discriminant becomes negative, no real equilibria exist, and all trajectories regardless of initial condition decline monotonically to zero, confirming that harvesting beyond the MSY inevitably leads to population extinction (Figure 3.5).

3.3 Logistic Growth with Periodic Harvesting

3.3.1 Model Formulation

In reality, fishing activity does not occur at a constant rate throughout the year. Fishing grounds are opened and closed according to seasons, weather conditions, and regulatory schedules. To model this more realistic scenario, the constant harvesting term a is replaced by a sinusoidal function that oscillates over time. The resulting periodic harvesting model is:

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) - a\,(1 + sin(bt)) \qquad \textbf{(3.8)}$$

The parameters appearing in equation 3.8 are described in the following table:

Table 3.3: Parameters of the Periodic Harvesting Model

Parameter	Description	Unit
P	Size of the fish population at time t	number of fish
t	Time	years or seasons
k	Intrinsic growth rate	per unit time
N	Carrying capacity of the environment	number of fish
a	Harvesting amplitude i.e. controls intensity of harvesting	fish per unit time
b	Angular frequency i.e. controls how often harvesting cycles occur	radians per unit time

The term $a\,(1 + sin(bt))$ represents a harvesting rate that oscillates between 0 and $2a$ over time. When $sin(bt) = 1$ the harvesting is at its peak intensity of 2a. When $sin(bt) = -1$ the harvesting drops to zero, representing a closed fishing season. When $sin(bt) = 0$ the harvesting equals a, representing the average rate. This structure naturally captures the opening and closing of fishing seasons throughout the year.

3.3.2 Role of Parameter *a*

The parameter a control the amplitude and average intensity of harvesting. It determines how much fish is removed on average per unit time and how large the seasonal swings are.

When a is small the oscillations are mild and the population recovers sufficiently between fishing seasons. As a increases toward MSY = $kN/4$ the peak harvesting rate reaches $2a$, which can temporarily push the population below its safe threshold. Even with partial recovery during the closed season, repeated exposure to high peak rates causes a net downward trend over time. This is why the effective safe harvesting limit in the periodic model is lower than the MSY derived for the constant model, the oscillations introduce additional stress that constant harvesting does not capture

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3.3.3 Role of Parameter *b*

The parameter b controls the frequency of the harvesting cycle. The period of one complete cycle is:

$$Period = \frac{2\pi}{b}$$

The following table presents the harvesting periods corresponding to different values of b, along with their biological interpretations:

Table 3.4: Harvesting Periods and Biological Interpretation for Different Values of b

Value of b	Period	Biological Meaning
0.5	12.57	Long slow cycles i.e. extended recovery periods
1.0	6.28	Standard seasonal fishing
2.0	3.14	More frequent harvesting i.e. less recovery time
4.0	1.57	Very frequent short seasons

When b is small the harvesting cycles are long and slow. The population experiences long periods of low harvesting during which it can recover substantially before the next peak. When b is large the harvesting cycles are short and rapid. The population is hit repeatedly with peak harvesting before it has time to recover, causing a stronger cumulative downward pressure on the population even if the average harvesting rate a remains the same.

3.3.4 Why No Analytic Solution Exists

The constant harvesting model $\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) - a$ is an autonomous ODE because the right-hand side depends only on *P* and not on time t. This allowed the equation to be solved analytically using separation of variables and partial fractions.

The periodic harvesting model $\frac{dP}{dt} = kP\left(1 - \frac{P}{N}\right) - a\,(1 + sin(bt))$ is a non-autonomous ODE because the sine term introduces an explicit dependence on time *t*. This has two important consequences.

First, separation of variables is no longer possible. When attempting to separate the equation:

$$\frac{dP}{kP\left(1 - \frac{P}{N}\right) - a\left(1 + sin(bt)\right)} = dt$$

the left-hand side contains both *P* and *t* mixed together through the sine term, making clean separation impossible.

Second, the concept of fixed equilibrium points breaks down. In the autonomous case *dP/dt* = 0 gave fixed values P_1 and P_2 that did not change with time. In the non-autonomous case setting

$\frac{dP}{dt} = 0$ gives: $kP\,(1 - \frac{P}{N}) = a\,(1 + sin(bt))$

The right-hand side changes with every value of t, meaning the equilibrium position itself shifts continuously over time. There are no fixed steady states, only time-varying quasi-equilibria that oscillate with the harvesting cycle.

For these reasons no closed-form analytic solution exists for the periodic harvesting model and the solution must be obtained entirely through numerical methods.

3.3.5 Results and Discussion

The following figures present the graphical results of the periodic harvesting model using the parameters k = 0.20, N = 5, and b = 1.0. Four figures are presented for two values of the harvesting rate a = 0.21 (below MSY) and a = 0.25 (at MSY) showing both the slope field and solution trajectories for each case. A fifth figure examines the effect of varying the frequency parameter b.

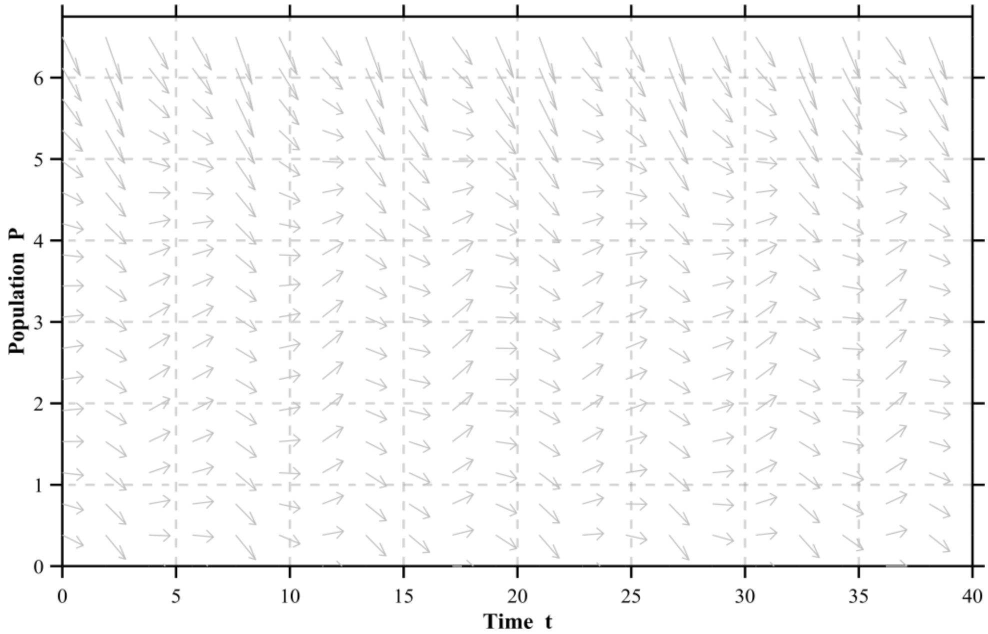

Fig 3.6: Slope Field for Periodic Harvesting ($a = a_1 = 0.21$)

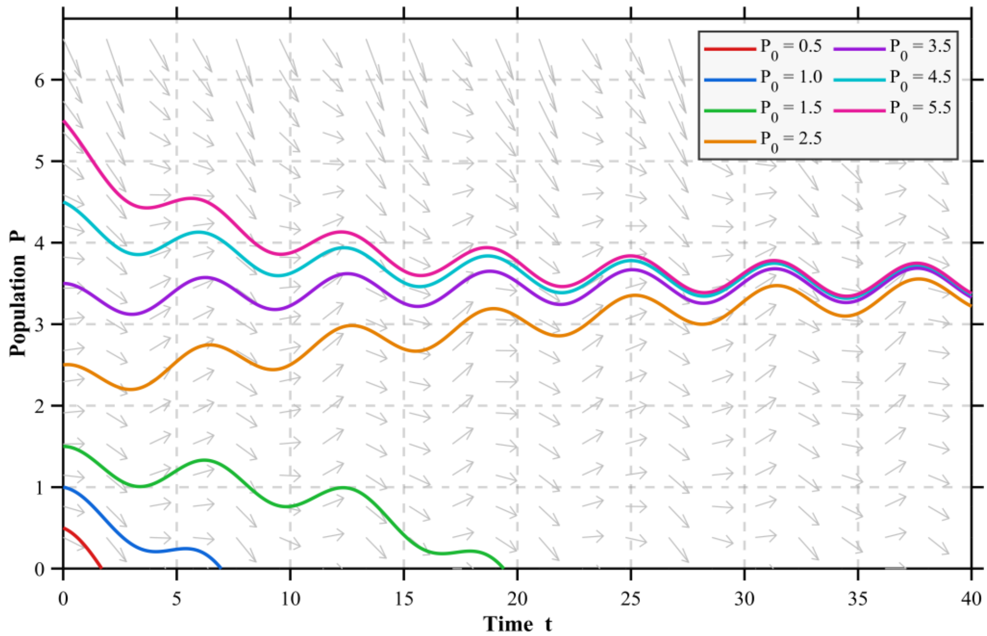

Fig 3.7: Trajectories for Periodic Harvesting ($a = a_1 = 0.21$)

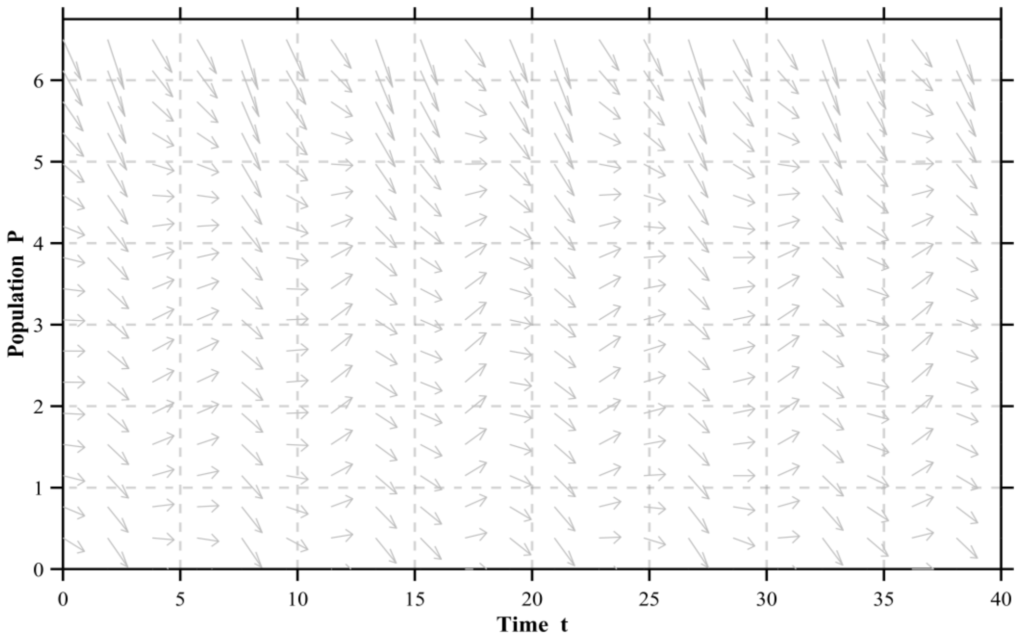

Fig 3.8: Slope Field for Periodic Harvesting ($a = a_2 = 0.25$)

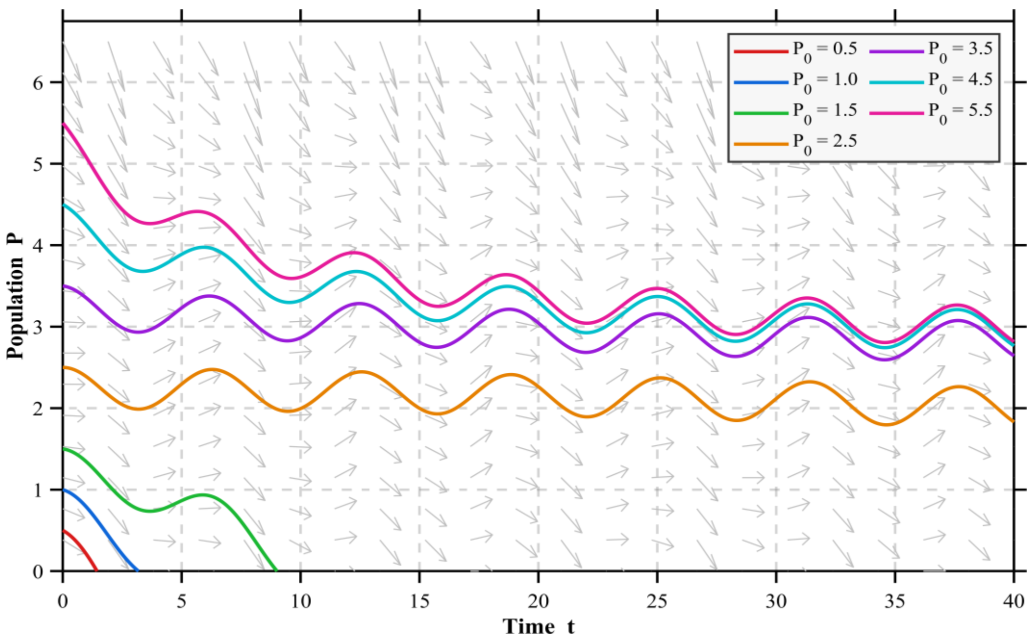

Fig 3.9: Trajectories for Periodic Harvesting ($a = a_2 = 0.25$)

The periodic harvesting slope field differs fundamentally from the constant harvesting case instead of uniform arrows at each population level, the arrows vary in direction across both P and t, reflecting the non-autonomous nature of the equation where the rate of population change depends on both the current population size and the current phase of the harvesting cycle (Figure 3.6).

For $a = 0.21$, trajectories starting above approximately $P = 1.5$ settle into sustained oscillations around a mean value below N, while those starting at $P_0 = 0.5$ and $P_0 = 1.0$ decline toward extinction during the early peak harvesting phase. Unlike the constant harvesting model, no fixed steady state exists, the population permanently cycles between a peak during the closed season and a trough during the open season (Figure 3.7).

At $a = 0.25$, the slope field shows more aggressively downward-pointing arrows, reflecting a peak harvesting rate of $2 \times 0.25 = 0.50$ double the MSY driving the population downward during peak harvesting periods (Figure 3.8). The trajectories still oscillate but with larger amplitude and a clear downward drift in the mean population level with each successive cycle, contrasting with the stabilized oscillations seen at $a = 0.21$. Low initial conditions reach extinction faster, demonstrating that setting the periodic harvesting amplitude equal to the constant model MSY is already unsustainable in practice (Figure 3.9).

3.4 Comparison and Sensitivity Analysis

3.4.1 Constant vs Periodic

To directly compare the two harvesting models, both are simulated using identical parameter values: $k = 0.20$, $N = 5$, $a = 0.21$, $b = 1.0$, and $P_0 = 4.0$. The only difference between the two simulations is the form of the harvesting term constant a versus periodic $a(1 + \sin(bt))$.

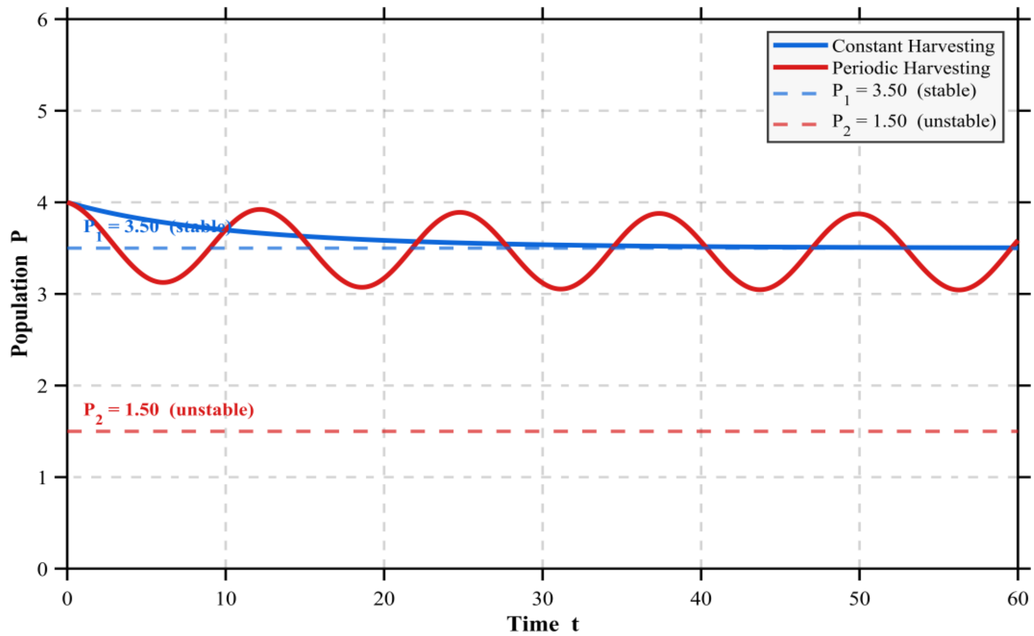

Fig 3.10: Constant vs Periodic Harvesting

The side-by-side comparison reveals a fundamental difference in long-term population behavior between the two models. Under constant harvesting the population follows a smooth monotonic trajectory that converges to the stable equilibrium P_1 = 3.5 and remains there indefinitely. The population settles into a fixed steady state with no further fluctuation. Under periodic harvesting with the same average harvesting rate a = 0.21, the population never reaches a fixed steady state. Instead, it oscillates continuously in response to the sinusoidal harvesting cycle, rising during low-harvest phases and declining during peak-harvest phases.

3.4.2 Which Model is More Realistic

From a biological and management perspective, the periodic harvesting model is more realistic than the constant harvesting model for the following reasons. Real fisheries do not operate year-round at a fixed rate. Fishing is governed by seasonal regulations, weather patterns, and migration cycles throughout the year. The sinusoidal function $a\,(1 + \sin(bt))$ captures this cycle naturally. In contrast the constant model a certain number units are removed per unit time regardless of actual population size, which is biologically impossible when fish are scarce. Furthermore, real fish populations do not sit at a fixed equilibrium. They fluctuate in response to environmental variability and seasonal productivity changes. The oscillating trajectories of the periodic model are conceptually closer to what is observed in real fishery data than the smooth convergence of the constant model. However, the periodic model also has limitations. A perfect sinusoidal harvesting pattern is still a simplification of irregular real-world fishing activity. Despite this, the periodic model provides a substantially more realistic framework for guiding fishery management decisions and should be preferred as the primary planning tool, with the constant model used as a theoretical reference.

The simulation results further reinforce this conclusion. The periodic model demonstrates that the effective safe harvesting threshold is lower than the theoretical MSY derived from the constant model, meaning that managers who rely solely on the constant MSY as their harvesting limit risk driving the population toward extinction without realizing it. The frequency parameter b adds an additional layer of control, allowing managers to simulate the effect of shorter or longer fishing seasons without changing the total harvesting intensity. A lower frequency corresponds to longer recovery periods between harvesting cycles, which the simulations show leads to more stable long-term population levels. In practice, combining a conservative harvesting amplitude below the constant MSY with a low harvesting frequency would represent the safest and most sustainable management strategy supported by this model.

3.4.3 Effect of Key Parameters on Population Dynamics

The following figures present the effect of four key model parameters — the harvesting frequency b, the intrinsic growth rate k, the carrying capacity N, and the harvesting rate a with parameters fixed at k = 0.20, N = 5, a = 0.21, b = 1.0, and P_0 = 4.0 in each case, varying only the parameter under investigation across both constant and periodic harvesting models.

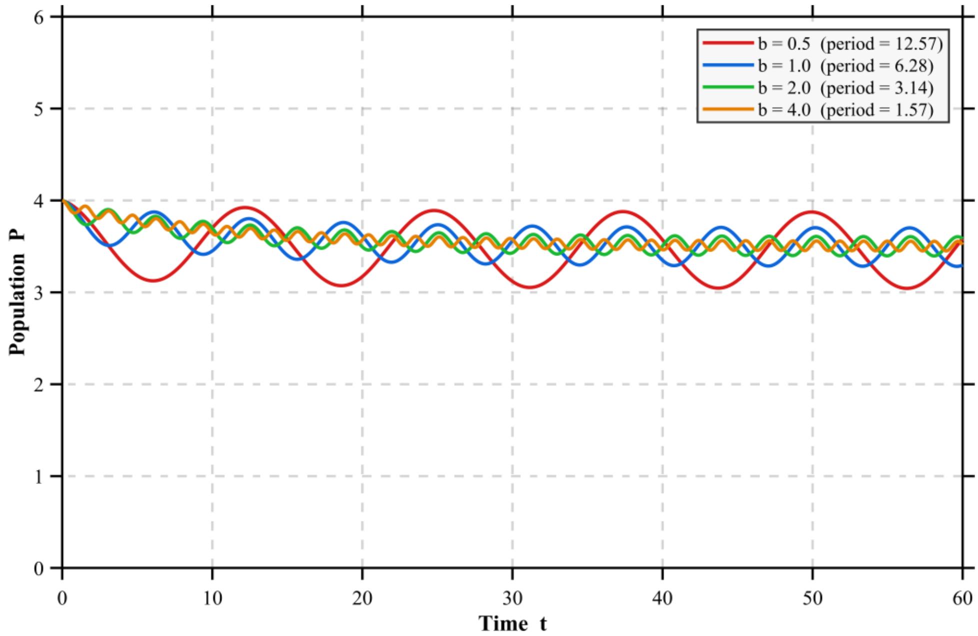

Fig 3.11: Trajectories for Varying *b* (*b* = 0.5, 1, 2, 4)

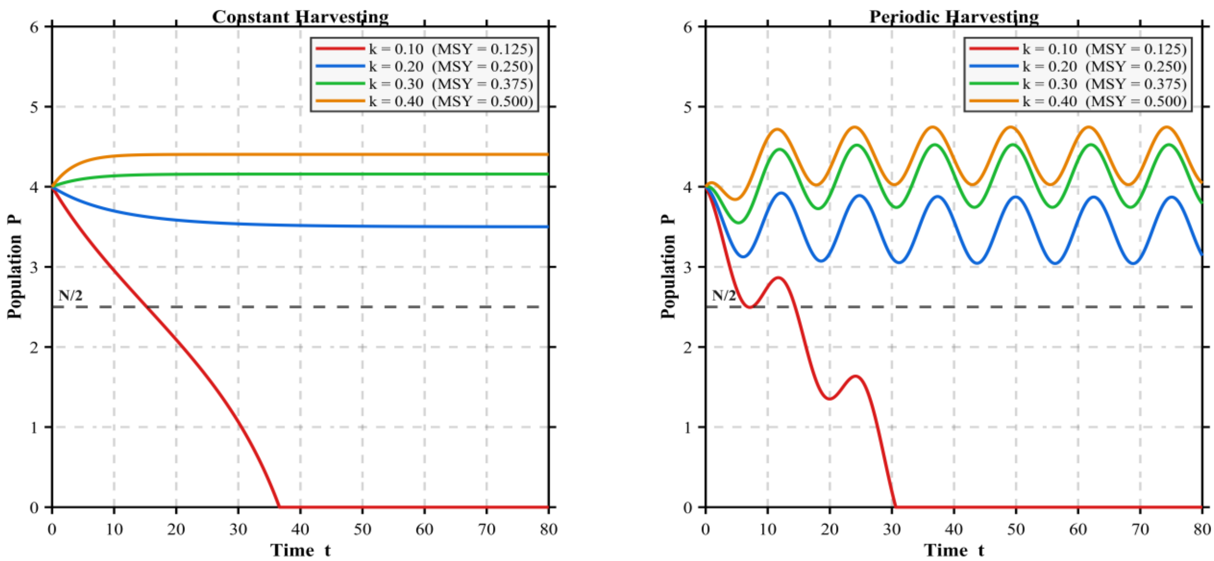

Fig 3.12: Effect of Varying *k* on Population Trajectories

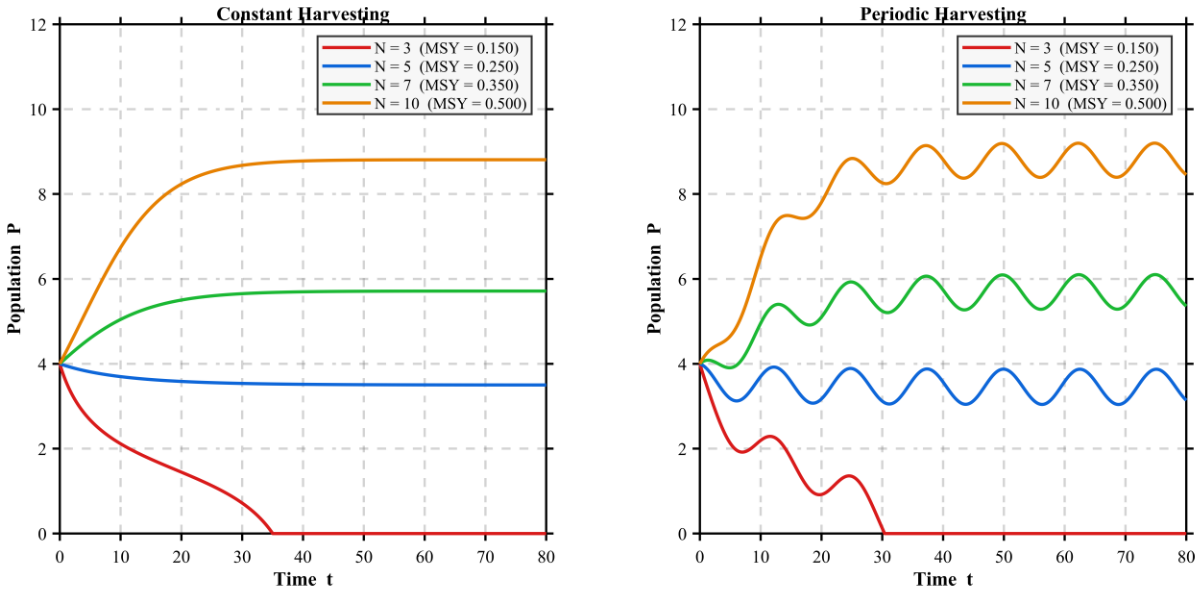

Fig 3.13: Effect of Varying *N* on Population Trajectories

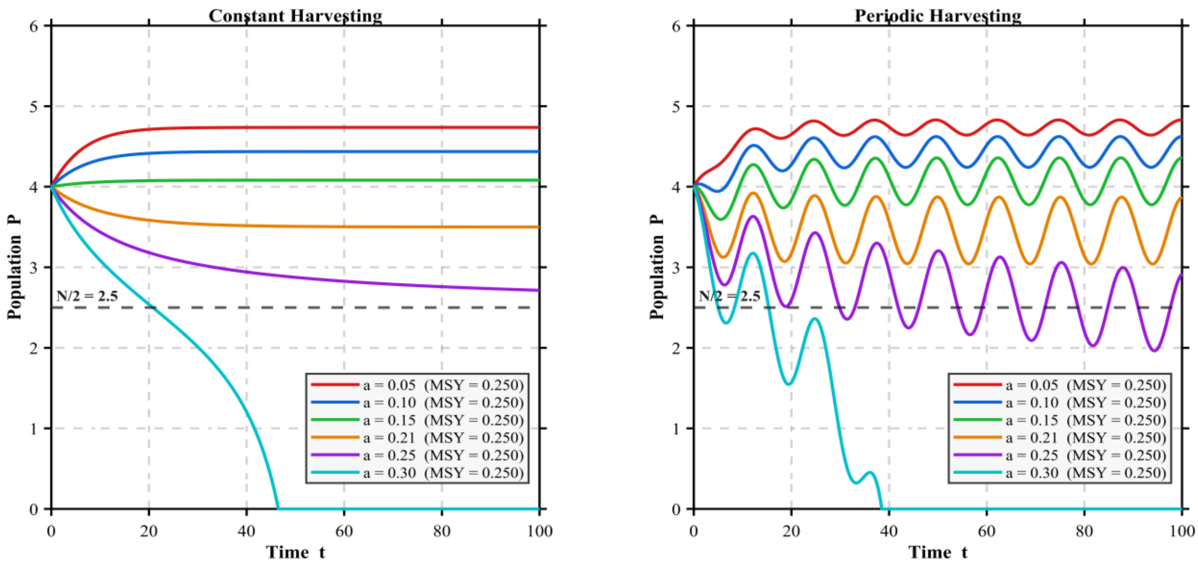

Fig 3.14: Effect of Harvesting Rate *a*

Figure 3.11 shows that all four trajectories maintain the same mean population level regardless of frequency, confirming that b does not affect long-run average behavior only the amplitude and speed of oscillations. Higher frequencies produce faster, smaller oscillations while lower frequencies produce slower, larger swings around the same mean. As k increases, the MSY increases proportionally, allowing the population to sustain higher harvesting pressure. At $k = 0.10$ the MSY falls below $a = 0.21$, driving the population to extinction under both harvesting modes, while for $k \geq 0.20$ the population stabilizes at progressively higher equilibria. Under periodic harvesting, surviving populations oscillate around mean values that closely match the constant harvesting equilibria, confirming that the constant model reliably predicts long-run average behavior (Figure 3.12). The same pattern holds for varying *N*, smaller carrying capacities result in lower MSY values that fall below the harvesting rate, causing extinction, while larger N values support higher stable equilibria under both models (Figure 3.13). Both harvesting models share the same collapse threshold at $a = 0.25$, but behave differently near it. Under constant harvesting populations converge smoothly to stable equilibria, while periodic harvesting produces lower mean populations with increasing oscillations as a approaches the MSY. When a exceeds 0.25 both models collapse to extinction, but periodic harvesting collapses more abruptly due to the additional stress of peak harvesting episodes (Figure 3.14).

The sensitivity of the equilibrium points to k is summarized in the table below:

Table3.5: Effect of Growth Rate k ($N = 5$, $a = 0.21$)

k	$\mathbf{MSY} = \frac{\mathbf{kN}}{\mathbf{4}}$	P_1 (constant)	P_2 (constant)	Status (constant)	Mean P (periodic)
0.10	0.1250	—	—	Extinction	Extinction
0.20	0.2500	3.5000	1.5000	Sustainable	3.50
0.30	0.3750	4.1583	0.8417	Sustainable	4.16
0.40	0.5000	4.4039	0.5961	Sustainable	4.40

The sensitivity of the equilibrium points to N is summarized in the table below:

Table 3.6: Effect of Carrying Capacity N ($k = 0.20$, $a = 0.21$)

N	$\mathbf{MSY} = \frac{\mathbf{kN}}{\mathbf{4}}$	P_1 (constant)	P_2 (constant)	Status (constant)	Mean P (periodic)
3	0.1500	—	—	Extinction	Extinction
5	0.2500	3.5000	1.5000	Sustainable	3.50
7	0.3500	5.7136	1.2864	Sustainable	5.72
10	0.5000	8.8079	1.1921	Sustainable	8.82

Chapter 4

Conclusion

This project explored how mathematical modelling can be used to understand bacterial infections, antibiotic resistance, and the role of the immune system. By developing and analyzing a system of differential equations, we were able to capture the complex interactions between sensitive bacteria, resistant bacteria, and immune response.

The study shows that the outcome of an infection depends strongly on the balance between bacterial growth, antibiotic effectiveness, and immune strength. In some cases, treatment can eliminate sensitive bacteria but still allow resistant strains to persist, while in others, both types of bacteria may continue to survive. Overall, the model highlights a key real-world insight: antibiotic treatment alone is not always sufficient, an effective immune response is equally important in controlling infections. This emphasizes the growing challenge of antibiotic resistance in modern healthcare.

In conclusion, mathematical models like this provide valuable tools for predicting infection behavior and can support better treatment strategies, while also pointing to the need for continued research in combating antibiotic resistance.

For the population model, logistic growth with harvesting showed that sustainability is highly dependent on the harvesting rate. A critical threshold exists, beyond which the population declines rapidly and may approach extinction, as observed near the limiting value of the harvesting parameter. Periodic harvesting, while more complex, provides a more realistic representation and can help regulate populations when applied appropriately. Numerical simulations and graphical methods further supported the analytical results by illustrating system behavior over time. Together, these models emphasize a common insight: maintaining balance within a system is essential for long-term stability.

In conclusion, this project highlights the power of differential equations as practical tools for modelling, prediction, and decision-making in biological and ecological systems, while also emphasizing the need for careful parameter control to ensure sustainable outcomes.

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